Available online at www.sciencedirect.com

ScienceDirect

journal homepage: www.elsevier.com/locate/he

Review Article

Heat transfer techniques in metal hydride hydrogen storage: A review

Mahvash Afzal, Rohit Mane, Pratibha Sharma *

Department of Energy Science and Engineering, Indian Institute of Technology Bombay, Mumbai, India

ARTICLE INFO

Article history:
Received 14 July 2017
Received in revised form
28 October 2017
Accepted 30 October 2017
Available online 22 November 2017

Keywords:
Hydrogen storage
Metal hydrides
Heat transfer

ABSTRACT

Metal hydrides have been under focus as a favorable hydrogen storage medium. The heat transfer to/from the metal hydride reactor bed is one of the major controlling parameters of the storage process. Consequently, a variety of heat transfer techniques have been employed till date to improve the system performance. In this review, an effort has been made to summarize the developments so far, assess the effectiveness of various heat transfer techniques and draw some inferences from the study which can contribute to a more effective design of heat transfer systems. Upon a comprehensive study of the existing literature a classification of heat transfer techniques was attempted and their relative effectiveness assessed with respect to system scale. It was observed that improvement of only thermal conductivity or heat transfer coefficient will not be able to improve system performance. An effective design should take into account the influence of both these parameters concurrently.

Contents

* *Corresponding author.*
E-mail addresses: pratibha_sharma@iitb.ac.in, pratibha.phy@gmail.com (P. Sharma).

https://doi.org/10.1016/j.ijhydene.2017.10.166

Nomenclature

c	Concentration, H/M ratio
M	Molecular weight, kg/kmol
P	Pressure, bar
T	Temperature, K
u	Velocity, m/s
R	Universal gas constant, J/mol/K
ΔS	Entropy of reaction, J/mol/K
ΔH	Enthalpy of reaction, J/mol
ε	Porosity
h	Heat transfer coefficient, W/m^2K
C_p	Specific heat capacity, J/kg/K
k	Thermal conductivity, W/m/K
Q	Heat source/sink, W/m^3

Subscripts

eq	equilibrium
f	final
e	effective
m	metal
g	gas
s	source/sink

Abbreviations

MH	Metal hydride
ENG	Expanded natural graphite
HTF	Heat transfer fluid
PCM	Phase changing materials
HPMH	High pressure metal hydride

Introduction

Hydrogen has a promising future as an energy carrier as it is renewable and has a high energy density. Although it is the most abundant element present on earth, very little of it (less than 1%) is present in molecular form [1]. Most of it is present bound to water or hydrocarbons. Also, being the lightest element, storing hydrogen in compressed or liquefied form is energy and cost intensive. Therefore, the utilization of hydrogen as an energy carrier involves challenges in production, circulation and storage. For storage of hydrogen, the use of metal hydrides offers a viable alternative due to reversibility of the process, lower storage pressure and high volumetric density [2]. However, the main impediment in the further development of metal hydride based hydrogen storage systems is their poor heat transfer characteristics [3,4]. In the present review, the various heat transfer techniques that have been employed in metal hydride hydrogen storage systems will be discussed.

Metal hydrides are compounds formed by the sorption of hydrogen into the host metal/intermetallic compound. Initially, hydrogen dissolves into the host metal forming a solid solution (α phase). When the hydrogen pressure and hydrogen concentration within the solid solution start increasing, the metal hydride formation starts (β phase) [1]. The thermodynamics of this process can be understood through a pressure composition isotherm (P-C-T). At equilibrium when both α and β phase coexist, the P-C-T curve shows a plateau. The absorption of hydrogen is an exothermic reaction, while desorption is an endothermic process. The former occurs only when the supply pressure is greater than the equilibrium pressure and the later occurs only when the pressure is lower than the equilibrium pressure. The reaction can be written as [5]:

$$M + x/2\ H_2 \leftrightarrow MH_x + \text{Heat}$$

When hydrogen is required, heat is supplied to the metal hydride bed and the reaction is reversed. Dissipation of the released heat and absorption of the supplied heat controls the chemical equilibrium and hence the rate of absorption/desorption of hydrogen. The magnitude of heat generated may be assessed from the fact that for a sample of ball-milled, doped MgH_2 2 kJ/mol of hydrogen have to be removed per second in order to keep the sample at a constant temperature [2]. Therefore, augmenting heat transfer rates is essential for improving the performance of a metal hydride storage system. Heat and mass transfer phenomenon within the metal hydride beds have been studied in great detail. Many review articles have discussed the development in metal hydride hydrogen storage systems, however, the previous studies have been more broad-based including various types of hydrogen storage techniques, reactor design and geometry etc. [6–9]. In this review it is aimed to present a focused study of the development of heat transfer techniques in metal hydride hydrogen storage systems.

Before proceeding with the heat transfer techniques developed in metal hydride hydrogen storage tanks, the governing equations for absorption/desorption in a metal hydride are presented in detail to give the reader an appreciation of the processes occurring within the reactor. These equations have been successfully used to model the behavior of hydrogen storage systems [10–14]. As heat is released upon the absorption of hydrogen, temperature of the metal hydride bed increases and so does the equilibrium pressure which is a function of temperature. The variation of equilibrium pressure is given by Ref. [15]:

$$P_{eq} = \exp\{\Delta S/R - \Delta H/(R^*T)\} \tag{1}$$

Where, ΔS and ΔH are the reaction entropy and reaction enthalpy respectively; R is the universal gas constant, T is the temperature. The absorption will not occur unless supply pressure is increased or the bed is cooled. Since, it is not possible to increase supply pressure infinitely, a heat withdrawal mechanism to remove the heat released can ensure that the reaction continues at a satisfactory pace. The equations which govern the heat transfer associated with absorption/desorption of hydrogen within the metal hydride bed are given by:

$$(\rho C_p)_e \frac{\partial T}{\partial t} + \rho_g C_{pg}(u.\nabla T) = \nabla.(k_e \nabla T) + Q_s \tag{2}$$

where,

$$(\rho C_p)_e = \varepsilon^* \rho^* C_{pg} + (1-\varepsilon)^* \rho_m^* C_{pm} \tag{3}$$

$$k_e = \varepsilon^* k_g + (1-\varepsilon)^* k_m \tag{4}$$

From the above we observe that the heat transfer characteristics in a metal hydride bed depend upon the specific heat capacity of the metal and the gas, the thermal conductivity of the metal and the gas and the source/sink term Q_S. In the above equations the effect of radiation has been ignored. However, it may be incorporated when dealing with high temperature metal hydrides such as MgH_2 [15,16].

Fig. 1 shows a simplified version of the heat conduction and convection to highlight the factors that govern a simple heat transfer process. A classification of reactors based on these factors follows. It can be observed that the heat transfer process is in a metal hydride bed depend upon the thermal conductivity of the material, the bed radius, the area across which heat transfer takes place, and operating parameters such as heat transfer coefficient 'h' and cooling/heating fluid temperature 'T_o'. Although a heat accumulation term exists in the case of metal hydride hydrogen storage, however, this is only a simplified exemplification.

On the basis of Fig. 1, any heat transfer improvement technique applied to the system may be classified as: a) improvement of thermal conductivity of the metal hydride bed which includes techniques like inclusion of metal foam and ENG in alloy bed b) addition of heat transfer area, accomplished through addition of fins, cooling tubes etc. c) improvement in operating parameters such as reduction in temperature of heat transfer fluid and increase in heat transfer coefficient, achieved by increasing mass flow rate of the heat transfer fluid. The classification of heat transfer techniques would be incomplete if the methods used for improving the system efficiencies were left uncovered. Therefore, these will be covered under the topic of non-conventional methods in the current review.

Fig. 2 shows the percentage of studies covering each of the different heat transfer mechanisms. It is evident that the use of cooling tubes and fins far outweighs any other known heat transfer mechanisms. Also from Fig. 3 it can be observed that lab-scale system studies far exceed the number of studies conducted on large–scale systems. Therefore, there is a dearth of data on how real life large scale hydrogen storage systems would perform. While studies on absorption and desorption are comparable but number of studies on absorption are more common than those on desorption. Some of the representative studies have been summarized in Table 1.

Most of the work done in the 1980s concentrated upon exploring the reaction kinetics of hydrogen, and absorption. It had already been realized that heat transfer to and from the reacting bed had a great bearing on the system performance. Efforts were on to improve the heat transfer by introducing foams, metal shavings etc. However, during this time the understanding of the system behavior was still limited. In the 90s and early 2000s, modeling and simulation of the hydrogen storage system became the major research area, as computational facilities became more accessible. System behavior was simulated with the help of 2-d and 3-d models which brought to light several aspects such as motion of the gas within the hydride bed, effect of expansion volume on system performance, effect of convection and radiation within the system etc. Since the late 2000s, tremendous exploration has been made possible due to availability of sophisticated modeling software. The focus has been on modeling various

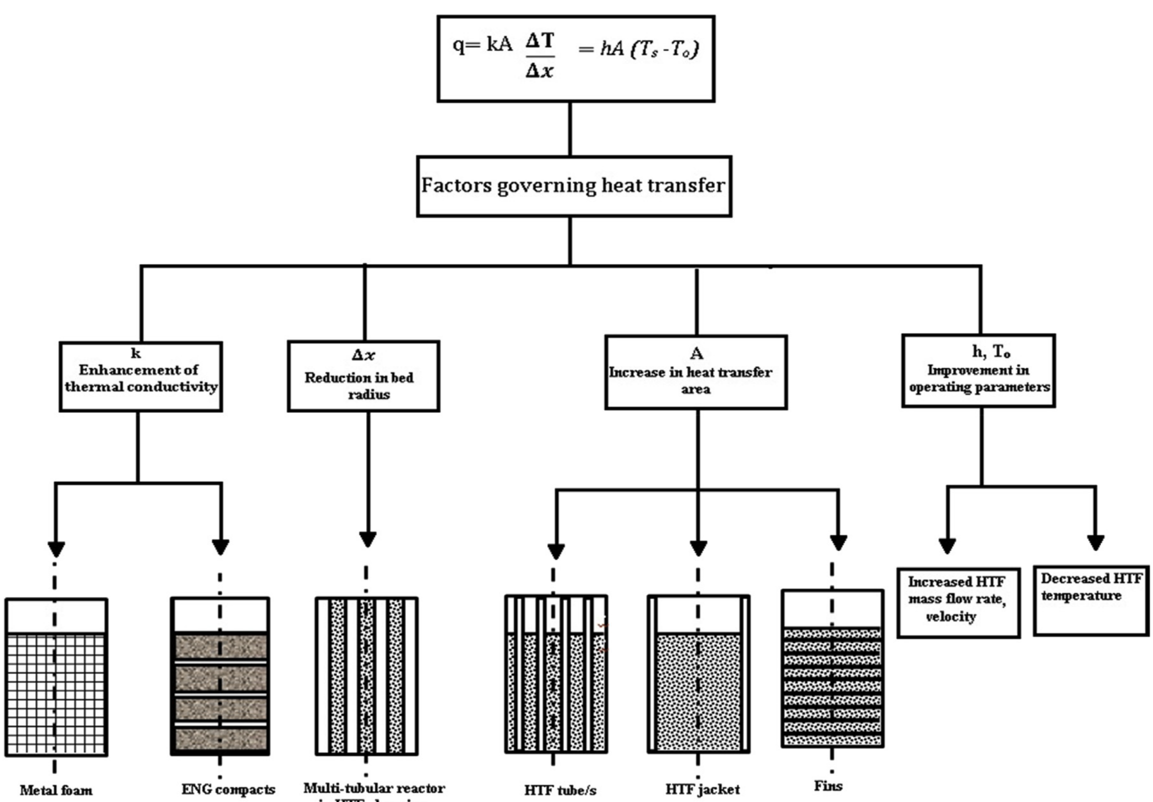


Fig. 1 – Classification of metal hydride reactors based on heat transfer parameters.

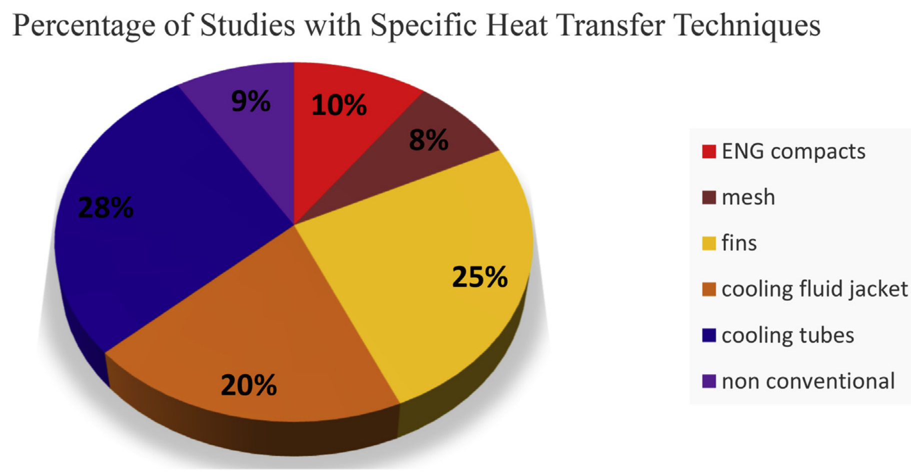


Fig. 2 – Comparison of heat transfer mechanisms studied.

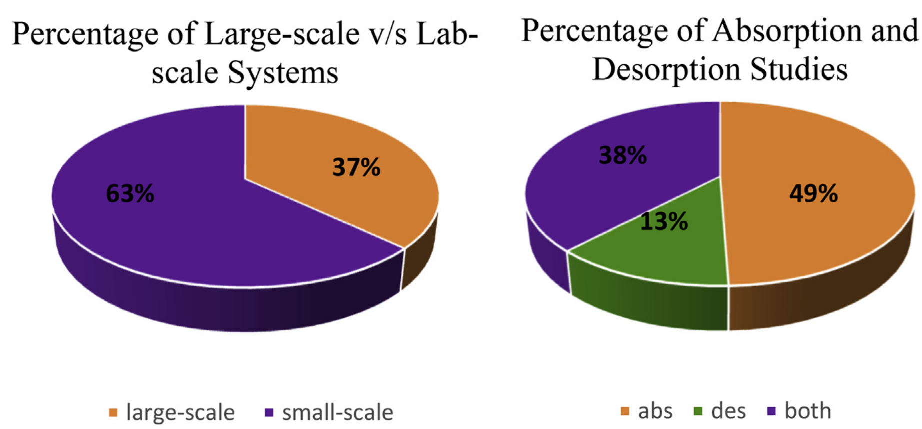


Fig. 3 – Comparison of systems studies based on scale and process studied.

types of reactor configurations, which would have been almost impossible through experimentation. A great variety of designs have been studied, optimized and compared. This has added vastly to the understanding of hydrogen storage in metal hydrides-reaction kinetics, heat and mass transfer in the reacting beds, system development etc. Reactor designs with optimized fin dimensions or optimum number of cooling tubes have been designed. Studies have explored the use of helical coils, helical coils with fins, heat pipes etc. In the subsequent section, we present a classification of various reactor configurations on the basis of the underlying heat transfer improvement as was explained in Fig. 1.

Improvement of thermal conductivity

A simple observation of the Fourier's law of heat conduction tells us that for improved heat conduction through any volume, a high thermal conductivity and low path radius are desirable properties. Choi and Mills who worked on metal hydrides for heat pump applications underlined the importance of bed radius as a deciding parameter for the addition of thermal conductivity enhancements [16]. Gopal and Murthy suggested that for effective heat transfer, the thermal conductivity of the bed should be maximized, while minimizing the radius [17–19]. However, the paradigm on the basis of which a bed radius could be decided remains, till date, very nebulous and varies according to the metal hydride and the reactor design. The main bottleneck, was nonetheless the internal thermal resistance of the hydride bed [4]. To this end, Nagel et al. built a network of copper wires which could be incorporated into the metal hydride bed [20]. The main idea was to come up with a cost effective method of improving thermal conductivity. However, the improved thermal conductivity was 0.4 W/m/K. As an alternative cost effective solution, Chen et al. investigated the effect of nickel foam on the performance of an AB_2 based metal alloy. They found that the thermal conductivity of the bed improved by a factor of 10. The foam structure also provided structural support to the powder and it was observed that the powder did not flow through the bed [21]. They also compared the results with non-continuous copper coated metal alloy powder compacts. The foam structure metal hydride bed exhibited better performance characteristics as compared to the copper coated alloy.

In 2007, Laurencelle and Goyette studied the effect of adding aluminum foam, shown in Fig. 4 (a), inside the reactor bed and experimentally validated the simulation results [22].

S. no.	References	Nominal dimensions (cm)	Materials	ΔH reaction J/mol of H_2	Thermal management system (heat transfer technique)	Pros	Cons
1	Z. Guo, 1999	N.A.	$LaNi_{4.7}Al_{0.3}$	−33819.7	Al plates (***Addition of heat transfer area***)	Improved heat transfer due to Al fins	Transverse fins may be subjected to cyclic stresses due to expansion/contraction while hydriding/dehydriding
2	Chen et al., 2002	18.5 × 10.5 × 10.5 and Φ6 × 33	$Ml_{0.85}Ca_{0.15}Ni_5$ and $Ti_{0.9}Zr_{0.15}Mn_{1.6}Cr_{0.2}V_{0.2}$	−26800 and −25400	Pellets with Cu coating and Ni foam (***Improvement of thermal conductivity***)	Ni foam proved to be a cost-effective method of improving thermal conductivity as compared to Al foam. Foam structure offers mechanical support to the alloy powder.	Pellets with Cu coating offer better volumetric efficiency
3	Sanchez et al., 2003	φ3 × 25	$LmNi_{4.85}Sn_{0.15}$	N.A.	Foam, graphite compacts (***Improvement of thermal conductivity***)	Compacts exhibit better heat transfer performance as compared to Al foam	Compacts may deteriorate over repeated cycling
4	Oi et al., 2004	10.5 × 21 × 32.6	Ti based AB_2 alloys	N.A.	Plate fin heat exchanger (***Addition of heat transfer area***)	Plate-fin heat exchanger design.	Stress concentration can occur at corners
5	Muthukumar et al., 2005	φ4 × 45	$MmNi_{4.6}Fe_{0.4}$ and $MmNi_{4.6}Al_{0.4}$	N.A.	Cu fins, water jacket (***Addition of heat transfer area***) (***Improvement in operating parameters***)	Effect of operating parameters observed: Inlet pressure most significant; heat transfer coefficient and heat transfer fluid temperature relatively insignificant	
6	Macdonald and Rowe, 2006	φ4 × 24	$LaNi_5$	−30,000	External fins (***Addition of heat transfer area***)	External fins improves the performance of the system up to 2/3 of reaction completion	Beyond 2/3 reaction completion external fins are not very effective in improving heat transfer
7	Laurencelle and Goyette, 2007	φ1.27 × 7.62	$LaNi_5$	−30500	Al foam (***Improvement of thermal conductivity***)	Importance of thermal conductivity enhancement established relative to system size	
8	Botzung et al., 2007	48 × 28 × 11.6	$MmNi_{5-x}Sn_x$		Al fins and foam (***Improvement of thermal conductivity***)	Combined Heat and Power system operating at low pressure and temperature (<3.5 bar and 70 °C)	
9	Forde et al., 2008	φ10.2 × 91.2	Ce modified $LaNi_5$	−27000	U-tube with fins (***Addition of heat transfer area***)	Integrated storage system with a PEMFC; waste heat from the fuel cell used for desorption process.	Different system required for absorption
10	Yang et al., 2008	–	$LaNi_5$	−31000	Tubular Reactor Disc Reactor Annulus Disc Reactor (***Improvement of thermal conductivity***) (***Addition of heat transfer area***) (***Improvement in operating parameters***)	System performance defined in terms of heat transfer controlled reaction rate and mass transfer controlled reaction rate. Heat transfer controlled reaction rate seen to be the controlling parameter.	
11	Askri et al., 2009	φ6 × 8	$LaNi_5$	N.A.	No enhancements, external fins, cooling tube, cooling tube with fins (***Addition of heat transfer area***)	Cooling tube with fins registered 80% improvement over no enhancement case	Addition of cooling tubes and fins affects volumetric and gravimetric capacity
12	Kaplan, 2009	φ2 × 12.2	$LaNi_5$	−30,478	External fins, water jacket (***Addition of heat transfer area***)	External water jacket more effective than external fins	

Table 1 – Summary of studies on typical heat transfer techniques.

(*continued on next page*)

Table 1 – (*continued*)

S. no.	References	Nominal dimensions (cm)	Materials	ΔH reaction J/mol of H_2	Thermal management system (heat transfer technique)	Pros	Cons
13	Freni et al., 2009	φ20 × 50	$LaNi_5$	−30800	Multiple cooling tubes, jacket (***Addition of heat transfer area***)	Thermal conductivity identified as a major heat transfer bottleneck External water jacket has higher efficiency than cooling tubes	
14	Ranong et al., 2009	Φ1.6 × 50 Φ6 × 100	$NaAlH_4$	–	Multitubular metal hydride tank Coiled cooling tube in single tank (***Addition of heat transfer area***)	Both designs exhibited comparable performances Coiled tube has better gravimetric capacity as compared to multitubular tank	Coiled tube was not chosen for scaling up; reasons were not explicitly discussed
15	Krokos et al., 2009		$LaNi_5$	−29879	Geometry optimization, HTF velocity optimization (***Addition of heat transfer area***) (***Improvement in operating parameters***)	System capable of on-board storage for upto 25 km	Volumetric/gravimetric penalties to be considered.
16	Linder et al., 2010	φ6.5 × 13.5	$LmNi_{4.91}Sn_{0.15}$	N.A.	Capillary tube bundle (***Addition of heat transfer area***) (***Improvement in operating parameters***)	Effect of reaction kinetics highlighted	
17	Dhaou et al., 2010	φ8 × 11	$LaNi_5$	N.A.	Spiral heat exchanger, with and without fins (***Addition of heat transfer area***)	Over 50% performance improvement relative to base case with no heat transfer enhancement.	
18	Mellouli et al., 2010	φ8 × 11	$LaNi_5$	N.A.	Double Spiral heat exchanger, with and without fins Optimization of pitch, fin diameter important (***Addition of heat transfer area***)	Double spiral heat exchanger with fins showed a 70% performance improvement over simple spiral heat exchanger	Additional volumetric/gravimetric penalties to be taken into account
19	Visaria et al., 2011	φ9.35 × 26.03	Ti-Cr-Mn	−14,390	HPMH system with fins, U-tube cooling (***Addition of heat transfer area***)	Optimized the design to achieve a fill time of less than 5 min for on-board storage systems.	An optimized 29% of the reactor volume was taken up by the cooling system.
20	Garrier et al., 2011	φ100	MgH_2	–	Tubes with internal fins and ENG (***Addition of heat transfer area***) (***Improvement of thermal conductivity***)	System energy density close to DOE targets	
21	Wang et al., 2012	r/h: 0.125 to 3.375	$LaNi_5$	−30000	Al foam (***Improvement of thermal conductivity***)		Effectiveness of fins against more diffused enhancements was not found to be promising.
22	Nyamsi et al., 2012	φ6	$LaNi_5$	31000	Cooling tube with fins (***Addition of heat transfer area***)	Fin effectiveness was seen to reduce with improved thermal conductivity	
23	Muthukumar et al., 2012	φ30 × 50	$MmNi_{4.6}Al_{0.4}$	−28,000	Multiple cooling tubes (***Addition of heat transfer area***)	Optimized number of tubes for a particular reactor design	
24	Garrison et al., 2012	max φ3.81	$NaAlH_4$	−84,000	Cooling tube with fins- transverse and longitudinal (***Addition of heat transfer area***) (***Improvement in operating parameters***)	Heat transfer performance for both types of fins was seen to improve	Volumetric and gravimetric penalties need to be compared

25	Visaria et al., 2012	φ10.6 × 35.5	Ti-Cr-Mn	−20700	Coiled tube heat exchanger (***Addition of heat transfer area***)		
26	Bhouri et al., 2012	φ6.5 × 13.5	$NaAlH_4$	N.A.	Multi-tubular reactor in a cylindrical shell (***Addition of heat transfer area***)	HTF temperature and velocity observed to impact system performance	The increase in fluid velocity only improved the performance up to a certain limit.
27	Delhomme et al., 2012	Φ13.8 × 90	MgH_2		Compacted Disc hydride with Expanded Natural Graphite (***Improvement of thermal conductivity***)	Hydrogen loading time of about 35 min was attained for a large scale reactor.	The changing crystal structure of the compacted discs mandates regular replacement.
28	Lozano et al., 2012	–	Sodium Alanate		Expanded Graphite (***Improvement of thermal conductivity***)	Minimum overall weight reactor design achieved while keeping constant predetermined performance parameters.	
29	Nakano et al., 2012	Φ16.52 × 100	$MmNi_5$		Double coil type heat exchanger (***Addition of heat transfer area***)	Large scale reactor successfully integrated with a Fuel Cell.	
30	Herbrig et al., 2013	φ4 × 13	N.A.	N.A.	(***Improvement in operating parameters***)	Parametric studies on systemic performance conducted	
31	Bao et, 2013	φ5 × 20	MgH_2	−75,000	Compacted discs with water jacket (***Addition of heat transfer area***) (***Improvement of thermal conductivity***)	Effect of operating pressures studied: inlet pressure, HTF velocity, HTF temperature.	
32	Chung et al., 2013	φ4.4 × 7.3	$LaNi_5$	−35,620	Heat pipe with fins ***(Non-conventional methods)***	50% improvement in the performance because of the use of heat pipes.	Similar performance improvement may not be true for large scale systems as the system considered was <22 mm
33	Meng, 2013	–	$LaNi_5$	−30000	Microchannel heat exchanger (***Non-conventional methods***)	Homogenous reaction attainment through critical adjustment of the center to center distance of the fluid channels.	Effects observed on very small scale system.
34	Garrier et al., 2013	φ13.8 × 100	MgH_2	−74000	MH compacts, PCM jacket (***Improvement of thermal conductivity***)	Employed PCM as thermal management option for large scale systems	PCM alloy reduced the gravimetric capacity and thus was recommended only for stationary applications.
35	Andreasen et al., 2013	Φ5 × 20	N.A.	N.A.	Internal and External Fins (***Addition of heat transfer area***)	Showed that internal fins are not effective when very low hydrogen discharge rates are required.	
36	Anbarasu et al., 2014	φ11.55 × 16	$LmNi_{4.91}Sn_{0.15}$	N.A.	Cooling tubes and jacket (***Addition of heat transfer area***) (***Improvement in operating parameters***)	Parametric study conducted; Impact of operating parameters was seen to depend upon reactor design.	
37	Wu et al., 2014	φ5 × 8	Mg_2Ni	−63,336	helical coil heat exchanger and straight tube heat exchanger (***Addition of heat transfer area***)	Helical heat exchanger was proved to be better than straight tube.	Stress analysis on helical systems over multiple cycles could be an issue.
38	Ma et, 2014	φ5 × 10	$LaNi_5$	N.A.	Cooling tubes with fins (***Addition of heat transfer area***) (***Improvement in operating parameters***)	Effect of fin number more significant than fin radius and thickness.	

(*continued on next page*)

Table 1 – (continued)

S. no.	References	Nominal dimensions (cm)	Materials	ΔH reaction J/mol of H_2	Thermal management system (heat transfer technique)	Pros	Cons
39	Ferekh et al., 2015	N.A.	ZrCo	N.A.	Foam, fins (***Improvement of thermal conductivity***)	Proved that foam based design superior to fins	
40	Lototsky et al., 2015	φ6 × 50	Ti based AB_2 alloys	−20000	Perforated Cu plate fins, ENG (***Improvement of thermal conductivity***) (***Addition of heat transfer area***)	Successfully integrated the system with an electric forklift	Though it was a large system, it did not purely use metal hydride as storage medium.
41	Boukhari et al., 2015	φ40	$LaNi_5$	−31,023	Multiple cooling tubes (***Addition of heat transfer area***) (***Improvement in operating parameters***)	Addition of cooling tubes improved the performance of the system.	Optimizing the number and dimensions of the cooling tubes is necessary to keep the weight of the system within acceptable limits.
42	Sekhar et al., 2015	φ6 × 50	$MmNi_{4.6}Al_{0.4}$	−28000	Straight tube, coiled tube, jacket, jacket with transverse fins (***Addition of heat transfer area***)	Compared different heat exchanger designs while taking into account the loss in volumetric capacity. Design with fins and external cooling most effective.	
43	Madaria and Kumar, 2016	φ3 × 13	$La_{0.8}Ce_{0.2}Ni_5$	−26730	Graphite flakes (***Improvement of thermal conductivity***)	Graphite flakes improved the system performance	
44	Kim and Kukkapalli, 2016	φ5	$LaNi_5$	N.A.	Peripheral longitudinal fins (***Addition of heat transfer area***)	Optimized the design for hydride tanks with internal longitudinal fins.	
45	Gkanas et al., 2016	φ10.51 × 40	Ti-Zr based AB_2; $LaNi_5$		Multiple cooling tubes with fins (***Addition of heat transfer area***)	Optimized number and dimensions of fins	
46	Tetuko, 2016	φ7.5 × 38	NA	N.A.	Heat pipes (***Non-conventional methods***)	Integrated a PEMFC with a hydride reactors using heat pipes.	Separate cooling mechanisms will have to implemented for absorption process.
47	Darzi et al., 2016	φ4 × 80	$LaNi_5$	N.A.	PCM jacket (***Non-conventional methods***)	Improvement in system performance due to metal foam in PCM jacket	Gravimetric penalties to be taken into account as PCM has high density.
48	Ben Maad, 2016	φ2 × 6	$LaNi_5$	−30932	PCM (***Non-conventional methods***)	Melting enthalpy of PCM is a crucial parameter along with thermal conductivity.	
49	Mellouli et al., 2016	max (φ40)	Mg_2Ni	−64000	PCM jacket, central tube, embedded units – spherical, cylindrical, hexagonal (***Non-conventional methods***)	Compared different PCM reactor designs and showed that cylindrical design performed better than the rest.	
50	Mellouli et al., 2017	φ4 × 16.43	Mg_2Ni	−64000	HTF tube and PCM material (***Non-conventional methods***) (***Addition of heat transfer area***)	Showed that HTF was more effective heat absorber than PCM over a larger time scale.	

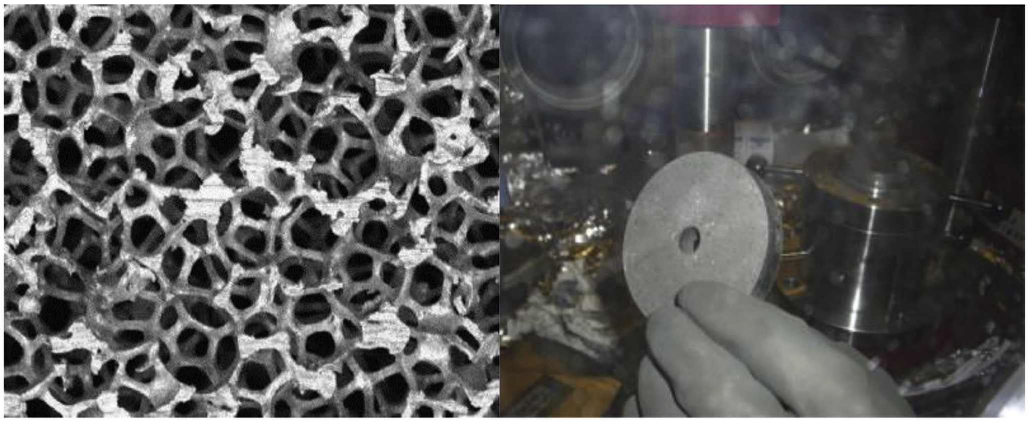

(a) (b)

Fig. 4 – a) Aluminum mesh for improvement of thermal conductivity [22] b) $NaAlH_4$ compacts [34].

They showed that a small reactor (diameter < 8 mm) may be operated without foam to complete the reaction in a certain interval of time whereas a larger reactor (diameter 6 cm) would require aluminum foam to achieve the reaction within the same time. Since improvements in thermal conductivity come at the cost of increased system weight and hence reduced overall gravimetric capacity, a need for determining the optimum value of thermal conductivity was recognized. Raju and Kumar modeled a high pressure metal hydride (HPMH), sodium alanate based system with an increased the thermal conductivity of a sodium alanate based on-board hydrogen storage system from 2 to 24 W/m/K [23]. In terms of weight fraction of hydrogen, the maximum performance was observed at 12 W/m/K. However, between 8.5 W/m/K and 12 W/m/K the performance did not show a significant improvement. This is indicative of the fact that thermal conductivity while being a major controlling factor is not the only factor. Similar results were also reported by Freni et al. who performed simulations on a $LaNi_5$ based cylindrical hydride system with multiple cooling tubes [24]. Reactor fill time for different values of bed thermal conductivity was calculated and a favorable fill time of 12 min was attained at conductivity of 10 W/m/K. Further increase in the value to 15 W/m/K, did not show any significant reduction in fill time.

An attempt to study the effect of thermal conductivity in combination with the kind of cooling environment, and the shape of the tank was made by Wang et al. [25]. They performed a parametric study using ANSYS Fluent, to quantify the effect of three parameters: increase in the effective thermal conductivity of the metal hydride, optimizing the shape of the tank, introducing an active cooling environment, on the performance of the tank. They observed that the aspect ratio of the tank is a critical parameter when the internal conductivity is low. However, generally conformability issues pose a constraint on the aspect ratio. It was also found that the amount of aluminum foam required to maximize performance under natural convection was higher than in case of active cooling on the outer surface.

Ferekh et al. studied the effects of internal heat transfer enhancements such as fins and metal foam [26]. The results demonstrated that the metal foam design is superior to the fin based design. It should be noted that these findings could be subject to the influence of operating conditions such as pressure and temperature and more importantly on the concentration of hydrogen absorbed/desorbed, as was also demonstrated by Madaria et al. [27]. Although mesh based systems seem to perform better as compared to fins, nonetheless they also reduce the volumetric and gravimetric capacity of the system which can be very crucial especially in case of mobile systems. In order to obtain a metal hydride with high thermal conductivity without much penalty of volumetric and gravimetric density, porous metal hydride compacts with high thermal conductivity metal additives were studied [28–31]. The idea was to reduce the gaps between particles and also add shavings of aluminum, copper, nickel to further increase the thermal conductivity of the compacts.

Sanchez et al. compared the conductivity values of $LmNi_{4.85}Sn_{0.15}$ hydride with graphite flakes and with aluminum foam [32]. Resulting values were in close agreement (approx. 19 W/m/K). A study by Pohlmann et al. revealed that the thermal conductivity could be increased to as high as 45 W/m/K at high compacting pressures [33,34]. Similar results were obtained for other materials lithium amide, sodium alanate, magnesium hydride and transition metal hydride Hydralloy C5 with expanded natural graphite. Although, this value may seem superfluously high, but as demonstrated by Dietrich et al. [35], the thermal conductivity reduces with cyclic usage. They observed that thermal conductivity reduced from 40 W/m/K to 12.5 W/m/K after 250 cycles. Beyond that it remained constant in the range of 10 W/m/K to 15 W/m/K. It should be noted that improvement in thermal conductivity was seen to be a strong function of ENG content and compaction pressure [34].

Bellosta von Colbe et al. scaled up the studies and experimentally investigated the performance of light weight $NaAlH_4$ based tank containing 4.4 kg of alanate [36]. The jacketed tank was filled with pelletized compacts as shown in Fig. 4(b), mixed with expanded graphite, resulting in improved thermal conductivity. The use of pelletized compacts was seen to be very effective in improving the performance, volumetric and gravimetric capacity of the system.

Addition of heat transfer area

Addition of fins

The addition of heat transfer area to the metal hydride system may be done through addition of cooling tubes or extended surfaces like fins. Nishizaki et al. as early as 1982, patented a reactor design with internal fins travelling from the center to the external heat transfer surface [37]. The fin design in this case was longitudinal. A transversal heat transfer surface study was conducted by Guo and Sung using aluminum sheets in the metal hydride bed via a 2-D mathematical model [38]. They demonstrated an improvement in heat transfer with use of aluminum plates and also found the optimum distance between two parallel plates. While most of the heat exchangers used are cylindrical in shape, Oi et al. demonstrated the enhancement in heat transfer characteristics using a plate-fin heat exchanger [39]. Botzung et al. improvised the plate fin heat exchanger by including aluminum foam with the metal hydride heat exchanger [40]. However, plate-fin heat exchangers may be subject to more failures due to stress concentration at corners. Also if untreated water is used as cooling fluid fouling may take place because of the small-sized channels.

Macdonald studied a metal hydride tank with external heat transfer enhancements and examined the case for natural and forced convection [41]. Upon a simple resistance network analysis, it was found that external fins had a significant impact upon heat transfer characteristics. However, once the reaction had proceeded to around two-thirds of completion the impact of fins reduced considerably. It may also be noted here that the diameter of the reactor was 4 cm. Therefore, while external heat transfer enhancements may be very effective for smaller vessels, the same may not be true when dealing with tanks of larger diameters. It would be worthwhile to mention here that use of fins is recommended when the heat transfer coefficient of the heating/cooling fluid is low, since the fin efficiency is an inverse function of the heat transfer coefficient. This implies that use of fins with high transfer coefficients could actually have an adverse effect of pointlessly increasing the conduction resistance if used for high heat transfer coefficients.

Following the same principal, Veerraju and Ram Gopal modeled the behavior of staggered air-cooled elliptical metal hydride tube bank with external plate-fins [42,43]. They estimated the eccentricity for maximum heat transfer. It was also observed that for similar cooling load, the elliptical tubes were more compact and used less fan power as compared to their cylindrical counterparts. However, while cylindrical tubes are available in a series of standard sizes, elliptical tubes will require specialized manufacturing.

As is the case with thermal conductivity, the number and design of fins also needs to be optimized keeping in view the overall system weight and volume. Visaria et al. developed a scheme for design of high pressure metal hydride reactors. They performed an optimization over the fin design. The optimum fin design reduced the absorption time to less than 5 min while occupying 29% of the reactor volume [44]. A comparative study of the effect of internal and external fins was performed by Andreasen et al. [45]. While internal fins improved the performance irrespective of the cooling fluid, the external fins registered insignificant improvement when the cooling fluid was water, a condition discussed earlier. A large scale hybrid hydrogen storage system was recently developed comprising of compressed H_2 and hydrogen stored in a finned AB_2 alloy [46]. They used a multi tubular system immersed in a water bath. Each tube of dimension 60 mm $\times$ 500 mm, comprises of perforated copper fins of thickness 0.5 mm at a distance of 5 mm. This system was used to operate an electric fork lift.

Askri et al. studied the impact of fin diameter upon the absorption of hydrogen [47] and observed that increasing fin diameter beyond a certain value had no effect on the hydrogen absorption. The effect of fin diameter was also studied by Kim et al. [48] and Melnichuk et al. [49]. Since fins improve the heat transfer characteristics by providing high thermal conductivity heat conduction path, fin arrangement within the reactor can also have a significant impact upon the performance of a reactor. Ma et al. studied the effect of fin configuration on the performance of a multi-cooling tube hydrogen storage system [50]. They noted that of all the parameters such as fin radius, thickness and number, fin number was most effective in improving the system performance. Similar results were reported by Singh et al. [51]. The use of fins in conjunction with ENG compacts was investigated experimentally by Garrier et al. [52] with a magnesium based air-cooled system. They reported a system energy-density of 270 Wh/kg, which was close to the DOE targets. A year later they improvised upon the design by using perforated fins and a cooling oil jacket [53]. The energy density improved to 360 Wh/kg. Although the difference in energy densities is huge it can partly be attributed to the fact that the former is a smaller system therefore the auxiliaries (sensors, heater etc.) account for nearly half the weight.

Nyamsi et al. performed an optimization study on a finned cooling tube in a metal hydride tank for desorption [54]. They observed that fin effectiveness decreased with increase in effective thermal conductivity of the metal hydride bed. Therefore, use of fins is not recommended in conjunction with metal foam or any other thermal conductivity enhancers. Kukkapalli and Kim studied the effect of longitudinal fins along the inner circumference of the metal hydride tank [55]. They optimized the number of fins and the fin aspect ratio for their reactor. Increase in number of fins augments the heat transfer surface area, so does the increase in fin diameter.

Addition of water jackets and cooling tubes

Although an absence of a unified design methodology for metal hydride reactors can be observed. Nonetheless, there have been attempts to optimize the designs for specific circumstances. A number of comparative studies on metal hydride tanks of varied geometries and designs have also been undertaken [56–62]. While many have underscored the best designs keeping in view the fastest absorption rate/highest reaction fraction, most have not constrained the same with system weight/volume, a condition that cannot be overlooked in real systems. Fig. 5 shows the comparative performance of some of the typical designs along with the materials used on

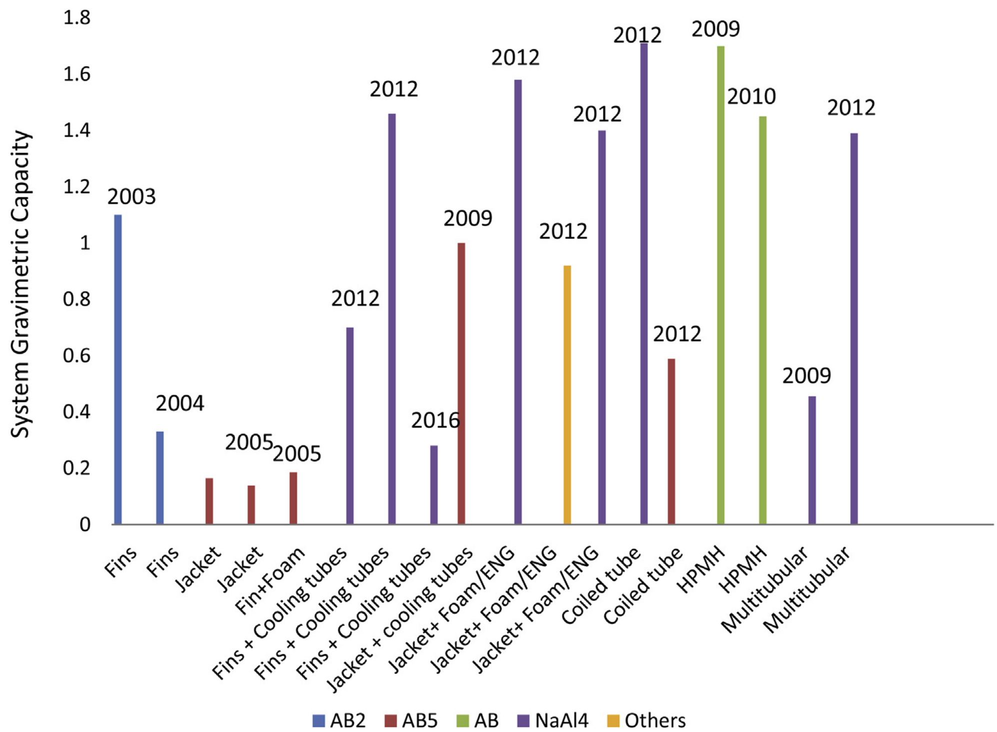


Fig. 5 – Comparison of system gravimetric capacity.

the basis of system gravimetric capacities. A majority of studies have not mentioned the system gravimetric capacity of the design which we believe is more significant since design of a hydrogen storage system comes with a weight and volume constraint especially in case of on-board storage. Even in stationary systems, most of the time there is a volume constraint in terms of stowing space available, ease of transportation for refilling etc. From Fig. 5 it is observed that sodium alanate systems seem to outperform others, which comes as no surprise since sodium alanate has a higher material gravimetric capacity than the others. HPMH systems and systems with coiled tube heat exchangers also appear competitive. It may also be noted that in case of fins and cooling tubes, the systems incorporating two or more types of heat exchange mechanisms seem to perform better than the systems with only either fins or jackets.

Askri et al. carried out a numerical analysis of four cylindrical metal hydride tanks as shown in Fig. 6: a) a base case tank that loses heat through its lateral and base surfaces b) tank with fins on its lateral surface c) similar to a) with the addition of a concentric heat exchanger tube d) similar to a) with the addition of a concentric heat exchanger tube with fins [47]. On comparing the designs it was found that the time required for 90% storage reduced remarkably. An improvement of 10%, 56% and 80% in the hydrogenation profiles over the basic case was registered. Kaplan performed an experimental study on three reactors at pressures varying from 1 to 10 bar [63]. The reactors tested were: a) simple metal hydride tank b) tank with twenty-two external fins c) tank with external water jacket. In all the three cases the dimensions of the metal hydride bed were same, diameter 20 mm and length 122 mm. It was found that the reactor with water jacket was the most effective in bringing down the temperature followed by the finned reactor. This was especially true when the supply pressure of hydrogen was higher. However, the improvements in these cases are highly dependent upon the tank size which was quite small (diameter < 5 cm) in both the cases.

C Na Ranong et al. performed a detailed analysis of two tank designs: a) single metal bed with embedded tube heat exchanger b) multiple metal tubes in a heat transfer fluid casing [64]. The latter performed better reaching 80% reaction completion in half the time as the former. However, the multi-tubular reactor was seen to have a much higher passive mass in the form of metal tubes fittings etc. Although, both the reactors had fill times falling within the desirable limits, the multi-tubular design was selected for manufacturing despite the high passive mass. The reasons for doing so were not made explicit. An optimization carried out by Lozano et al. attempted to optimize the tank diameter while minimizing the system weight [65]. More complex systems are difficult to optimize because there are too many control variables in the picture. There is the number and dimension of fins and cooling tubes, the porosity of foam or content of ENG whichever is being used, the diameter of the tank or number thereof in case of multi-tubular elements. Therefore, most studies concentrate on the optimization with respect to a single variable.

Upon increasing the concentric cooling tube diameter, about 25% improvement in performance was noted by Nyamsi et al. [54]. Since multiple cooling tubes of a smaller diameter could provide a greater and more uniformly distributed heat transfer surfaces as compared to a single concentric tube, therefore, independent studies to observe the effect of multiple cooling tubes through the metal hydride bed were conducted

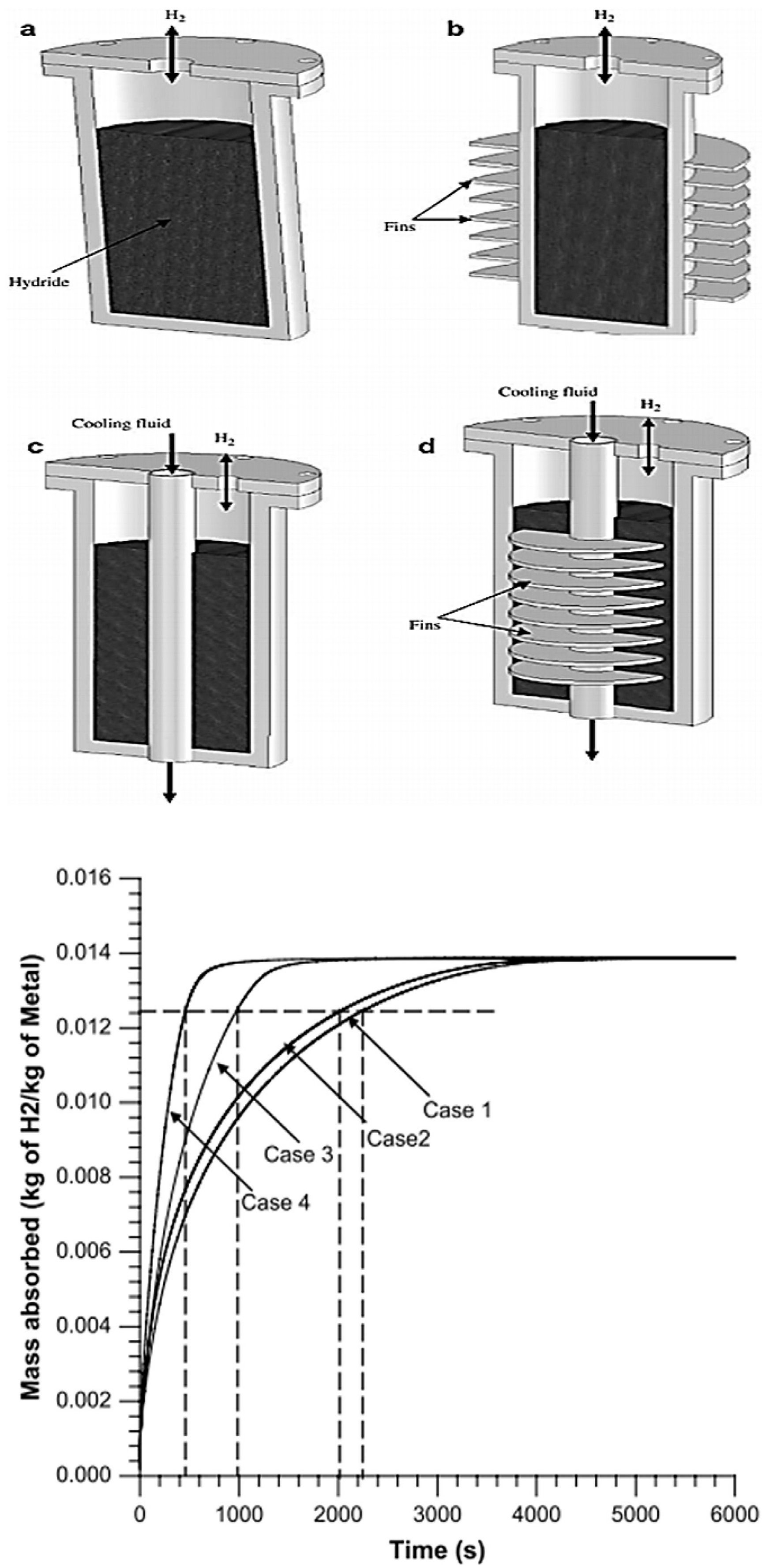
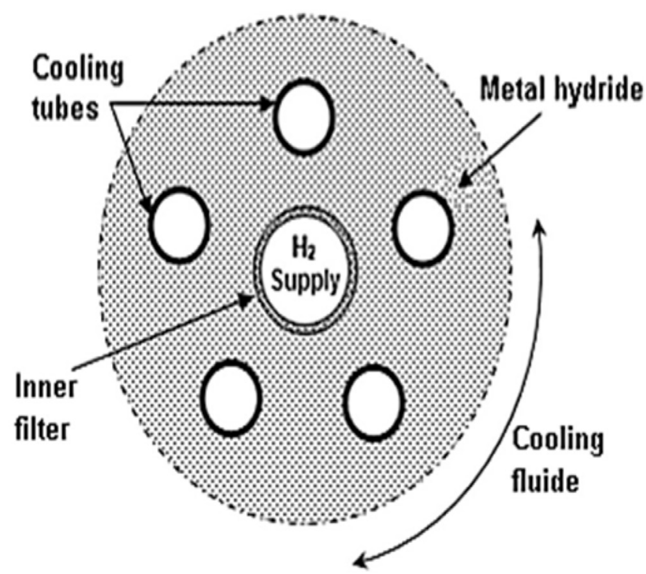
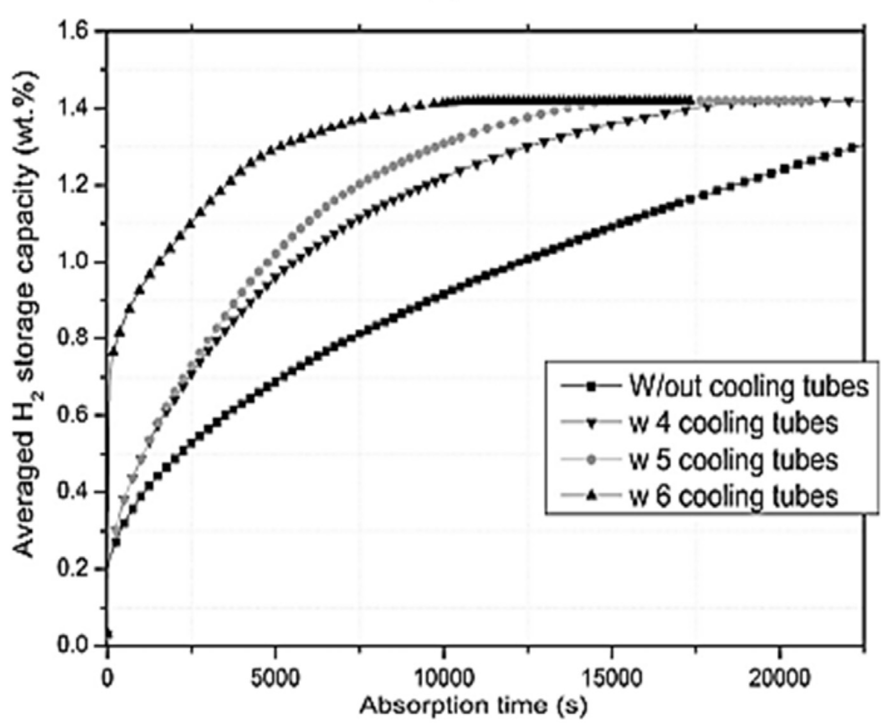


Fig. 6 — Comparative study of four types of metal hydride reactors [47].

[66,67]. The sectional view of the system used by Boukhari et al. and influence of number of cooling tubes on system performance are depicted in Fig. 7(a) and (b) respectively [67]. It was found that while including multiple cooling tubes greatly expedited the heat transfer process, the dimension and number of the cooling tubes however, must be optimized to avoid superfluous system weight. Bao et al. optimized the parameters including number of cooling tubes, tube diameter and pitch using Taguchi method [68]. They observed that the most important parameter affecting the heat transfer was the number of cooling tubes. Increasing the number of cooling tubes increases the number of high reaction surfaces by ensuring maintenance of low temperature at multiple locations, as lower temperatures result in faster reaction.



(a)



(b)

Fig. 7 – (a) Sectional view of metal hydride tank (b) Improvement in hydrogen storage with addition of cooling tubes [67].

Freni et al. also studied the impact of embedded cooling tubes on metal hydride systems [24]. They reported that while cooling tubes improved the absorption characteristics, in all cases they were limited by the low thermal conductivity of the metal hydride bed. They also reported that the external cooling jacket demonstrated a higher cooling efficiency. It remained to be seen whether the position of cooling tubes had any impact on the system performance. Muthukumar et al. numerically investigated the performance of an industrial scale system with multiple cooling tubes [69]. The number of cooling tubes and their location in the metal hydride bed were optimized. The optimum position of cooling tubes varies according to system size, configuration etc. Therefore it is different for different systems. It should be kept in mind that provision of extra heat transfer areas and improvement of thermal conductivity are two different aspects of the heat transfer problem, neither can make up for the deficiencies of the other. Therefore, in conjunction with cooling tubes there should always be high thermal conductivity pathways leading to the tube. These may either be in the form of fins, mesh or Expanded Natural Graphite. In the latter case the penalty in terms of weight and volume is the least with high returns in terms of conductivity.

An improvement over the cooling tubes, are cooling tubes equipped with fins. A few points to be noted when using transverse fins could be the problems associated with refilling the tank. Also, due to the expansion and contraction in the powder there could be structural damage to the fins. Garrison et al. performed the optimization of a $NaAlH_4$ based hydride tank [70]. They studied the effect of longitudinal fins versus radial fins both attached to a cooling tube. Both the designs were found to be equally effective with the longitudinal finned cooling tube performing slightly better than the transverse one. Gkanas et al. also studied the optimization of fins in a tank with multiple cooling tubes [71]. Cooling tube without any fins, with radial fins and longitudinal fins were studied. The response of each reactor configuration to different alloy families was noted. They observed that longitudinal fins presented a better performance with AB_2 alloy in comparison to $LaNi_5$. The number of fins, fin length and thickness were also varied. Optimized results were obtained for $LaNi_5$ and an AB_2 type alloys.

In order to benefit from a larger heat transfer area, Mellouli et al. developed a spiral heat exchanger [72] and furthered the study by studying different configurations of the heat exchanger including spiral heat exchanger with fins, double spiral heat exchanger, double spiral heat exchanger with fins [73]. They also studied the impact of fin length, thickness and pitch on the heat transfer. However, the improved heat transfer characteristics come at the cost of system size and gravimetric capacity. To overcome these issues and gain an added advantage of shorter fill times, High Pressure Metal Hydrides (HPMH) undergoing absorption at 350–700 bar, were studied to cater to the requirements of fuel cell vehicles [74,75]. Visaria et al. presented a detailed design methodology for designing a heat exchanger for onboard storage of HPMH [76–78]. Different configurations with a coiled tube and U-tube heat exchangers were studied. The U-tube heat exchanger was observed to satisfy the U.S. Department of Energy fill time condition for high pressure metal hydrides (280 bar).

The spiral heat exchanger was studied by Wu et al. in comparison to the single tube heat exchanger [79]. The former was found be more effective than the single tube heat exchanger. Although coiled tube heat exchanger has been successfully demonstrated for large scale systems also [80,81] but the response to stress for such systems over multiple cycles needs to be studied. Sekhar et al. compared the designs of different heat exchangers taking into account the loss in gravimetric capacity as well [46]. With a height to diameter (h/d) ratio of over 8, the authors found the effects of external cooling superior to internal heat exchangers (both straight and coiled) for the metal hydride tank of diameter 6 cm.

Improvement in operating parameters

Other than the design of the reactor, the operating conditions of the heat transfer fluid can have a great bearing on the reactor performance. In this section the effects of the operational parameters such as heat transfer fluid velocity, temperature etc. are explored. It is expected that increase in heat transfer fluid (HTF) mass flow rate will result in an improved performance due to higher Reynolds' number and hence better heat transfer. What remains to be identified is how much of it is beneficial in a practical system. Correspondingly, decrease in HTF temperature will improve the heat transfer by Newton's law; however, if it is worthwhile to add the costs and weight of a cooling system needs to be seen. Muthukumar et al. studied the absorption and desorption in misch-metal based alloys with change in operating parameters [82]. They observed that the decrease in heat transfer fluid temperature resulted in an increase in heat transfer for different alloys, however, the extent of the influence varied. In another study by the same group, it was observed that in comparison to the effect of increasing the supply pressure, the impact of increasing the heat transfer coefficient and the cooling fluid temperature is very insignificant, especially during the first 20 s of the reaction [83,84]. Similar results were also reported by Ye et al. [85].

Effect of heat transfer fluid velocity was also studied by Krokos et al. and found 1 m/s as the optimum value for that particular configuration [86]. The study was conducted on an optimized multitubular arrangement (multiple metal hydride reactors in HTF housing). Gambini et al. developed a lumped model to study the performance of metal hydride bed as a part of a larger fuel cell based system [87]. Their system comprised of a misch metal based reactor with an external water jacket and eight internal copper fins. They investigated the absorption and desorption behavior of the metal hydride with variation in heat transfer fluid (HTF) temperature and flow rate to ensure a smooth coupling with a fuel cell. Bhouri et al. also studied the improvement in hydrogen absorption under the influence of heat transfer fluid temperature and velocity [88]. It was observed that increasing the velocity of heat transfer fluid up to 2 m/s resulted in a significant improvement in hydrogen absorption; however, increasing the velocity beyond

2 m/s, the bed did not seem sensitive to the change. This finding was corroborated by Bao et al. [89]. The effect of velocity and temperature of the heat transfer fluid for a MgH_2 reactor was observed as shown in Fig. 8. The hydriding time was seen to decrease with increased heat transfer fluid velocity because of the increase in the heat transfer coefficient. However, beyond 5 m/s the effect becomes nominal as the thermal conductivity of the hydride bed becomes the

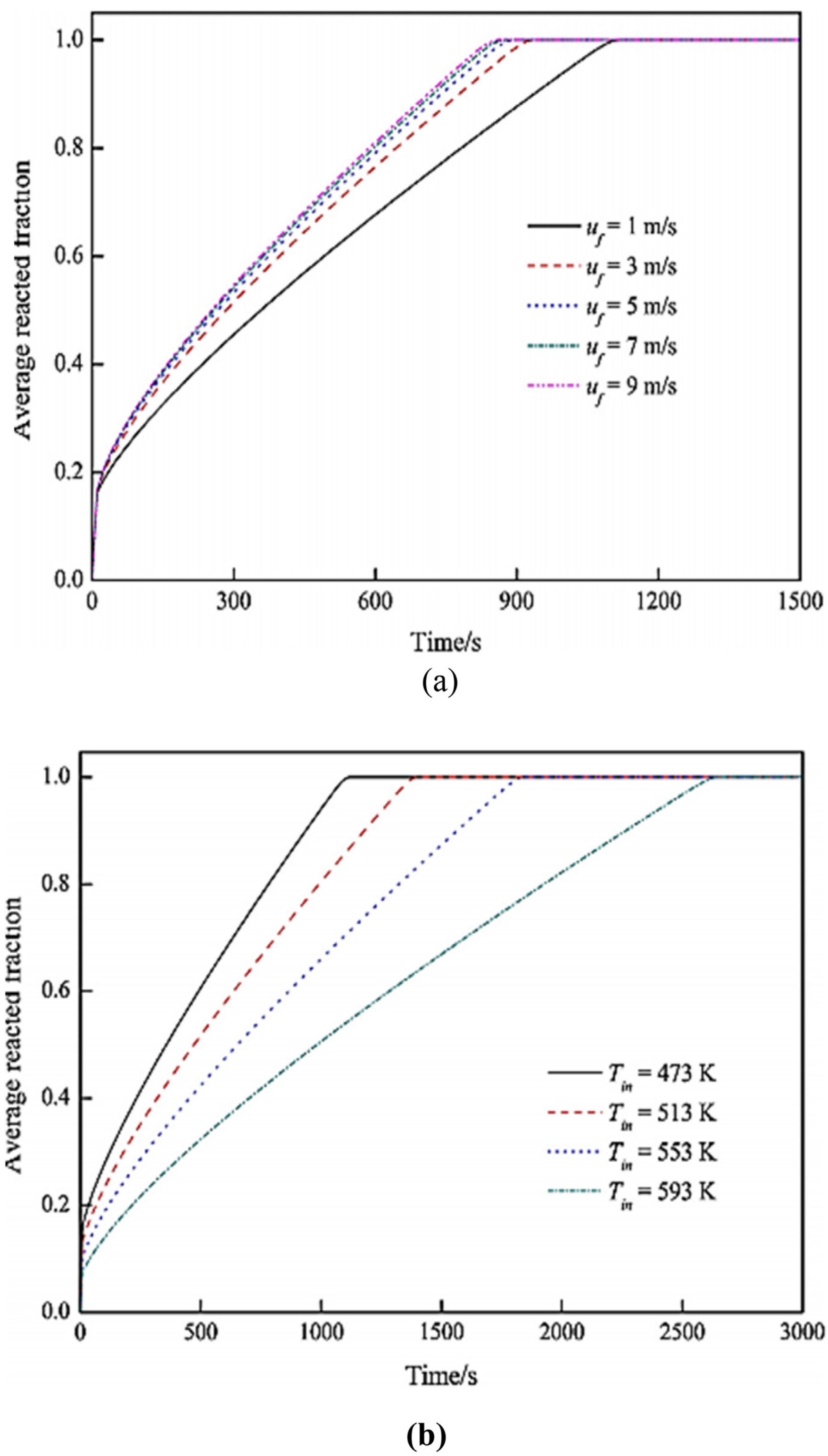


Fig. 8 – Effect of (a) HTF velocity and (b) HTF inlet temperature on amount of hydrogen absorbed [89].

dominant controlling parameter. Anbarasu et al. experimentally studied the effect of operating conditions on hydrogen absorption and desorption [90,91]. They noted that with increase in flow rate of cooling fluid, the rate of hydrogen absorption also improved, however it was highly dependent upon the reactor design (number of cooling tubes in their case).

Valizadeh et al. examined the performance of a metal hydride using the Lattice Boltzman method, substantiating the previous studies on effects of heat transfer fluid temperature and heat transfer coefficient [92]. However, from the study of various available reports it can be safely said that any improvement in heat transfer coefficient or reduction in heat transfer fluid temperature will bear any significant result only if the thermal conductivity of the metal hydride bed is simultaneously improved.

To study the effect of reaction kinetics upon the absorption and desorption of hydrogen Linder et al. considered a $LmNi_{4.91}Sn_{0.15}$ based capillary tube reactor with heat transfer fluid flowing through multiple capillary tubes [93]. The flow rates were varied from 2 l/min to 5 l/min. They observed that while for absorption improving the fluid flow rates improved the hydrogen absorption rate but for desorption the same could not be said. This implies that other than the limitation of heat transfer within a metal hydride bed, there could be an intrinsic reaction kinetic limitation.

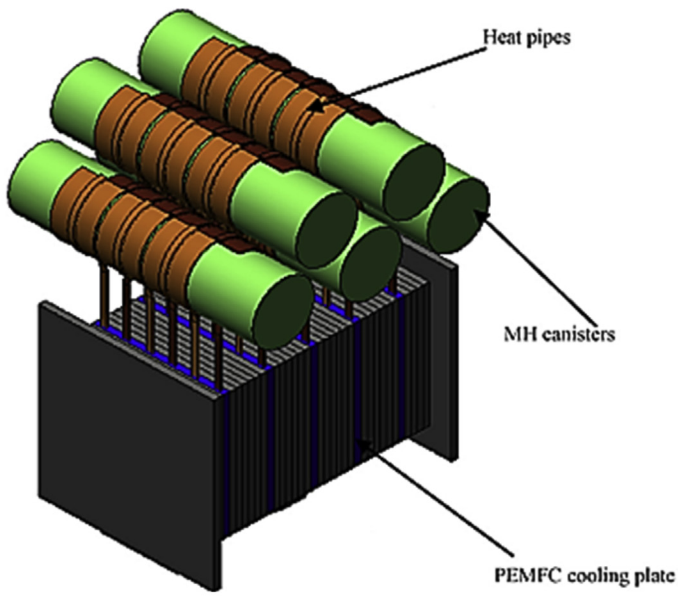
Yang et al. performed an interesting analysis through which they evaluated the heat transfer limited reaction rate and the mass transfer limited reaction rate for three different reactors: tubular, disc, and annular disc reactor [94]. In all three cases it was found that heat transfer limited reaction rate was lower than mass transfer limited reaction rate. Upon improvisation of the model, they found that the absorption reaction could be divided into two phases a mass transfer limited initial phase and a heat transfer limited second phase [95]. Borzenko et al. divided the hydrogen absorption in a metal hydride into three stages [96,97]. At the beginning of the absorption reaction, the inlet hydrogen pressure is above the equilibrium pressure, consequently hydrogen is easily absorbed in the hydride bed. Since the absorption is very intensive in this stage, and the metal hydride has an inherently low thermal conductivity the cooling mechanism is unable to completely remove the heat generated. This halts further absorption of hydrogen into the bed and results in a heat and mass transfer crisis. A similar situation is encountered during the discharge process. They proposed that a regulated volume flow rate of hydrogen could help in averting the heat and mass transfer crisis and improving heat transfer. Herbrig et al. presented a numerical model to study the dynamics of hydrogen storage bed with pelletized hydride-graphite composite [98] and followed it with an interesting analysis, strengthening the premise that the study of heat transfer processes cannot be mutually exclusive. They showed that hydrogenation process has three stages. In the initial stage, hydrogen mass transfer was found to be the dominant factor. Thermal conductivity of the bed governed the rate in the intermediate stage. Finally, heat transfer was found to dominate the overall process.

Non-conventional methods

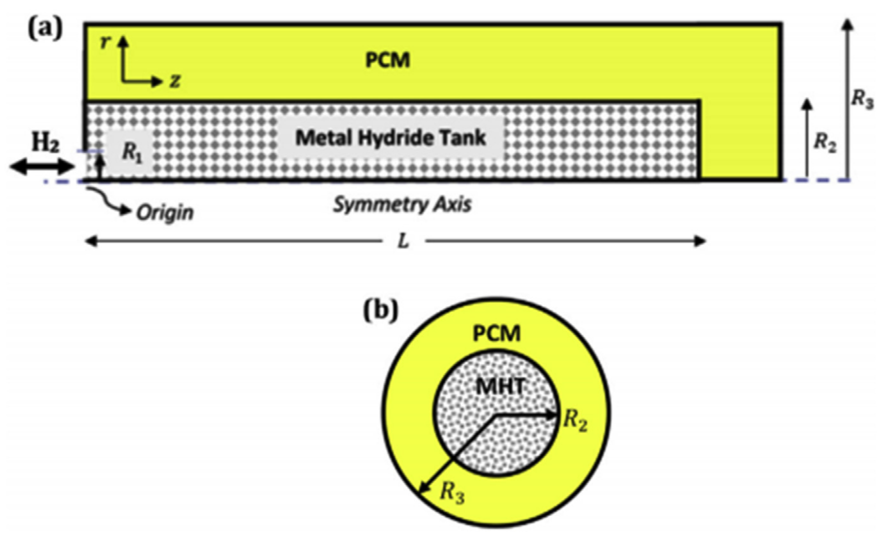
Recently, there have been many innovations in the design of metal hydride reactors including the development of micro channel reactors, heat pipe and the use of phase changing materials (PCM). Of these, the use of heat pipes and PCM lead to an improved system efficiency while they may or may not be competitive on the performance front. Førde et al. employed a U-shaped copper tube with fins to enhance the heat transfer characteristics of a metal hydride system integrated with a PEM fuel cell [99]. The waste heat from the fuel cell was used for desorption of hydrogen. This method could be useful for a metal hydride which desorbs hydrogen close to room temperature and could improve the system efficiency. Chung et al. used heat pipes for heat transfer in both absorption and desorption [100,101]. They observed a 50% improvement in the absorption and desorption behavior of the hydride. The diameter of the reactor was however 22 mm, making heat pipes a suitable heat transfer mechanism due to the small size.

Tetuko et al. modeled a coupling of a 500 W PEM fuel cell with metal hydride reactor through heat pipes, creating a passive system as shown in Fig. 9(a) [102]. Five metal hydride reactors of diameter 75 mm and length 380 mm were used. Five heat pipes attached to the metal hydride reactors were used for removing 880 W for a hydrogen release rate of 2.5 SLPM. It may however, be kept in mind that while this method is promising from a desorption perspective the system will have to be provided with a separate heat withdrawal mechanism for absorption. Meng et al. developed a mini channel heat exchanger for a rectangular metal hydride tank [103]. They observed that the heat transfer was a strong function of the center to center distance between the fluid channels. The reaction front was observed to be more homogenous due to the uniformly distributed channels. Mudawwar and Visaria patented a metal hydride tank which consisted of a modular heat exchanger where each module included a micro-channel heat exchanger while a coiled tube was used to cool the overall system [104].

Garrier et al. employed PCM to improve the heat transfer in a large scale MgH2 tank [105]. It was noted that due to the weight of the PCM alloy, the system gravimetric capacity was rather low. However, they still recommend it as a viable option for large scale stationary applications. Darzi et al. studied the use of Rubitherm phase change material jacket of radius 7 cm around a $LaNi_5$ tank of radius 4 cm as shown in Fig. 9(b) [106]. They observed that including metal foam in the PCM jacket, greatly improved the system performance. Ben Maad et al. also studied numerically the performance of a metal hydride system equipped with a PCM jacket [107]. They examined the effect of thermal conductivity and melting enthalpy of the PCM on the system performance. Melting enthalpy was found to be a crucial parameter governing the heat transfer while increase in thermal conductivity up to a certain value improved the system performance; any improvement beyond that value did not result in any significant changes. In some of the studies on



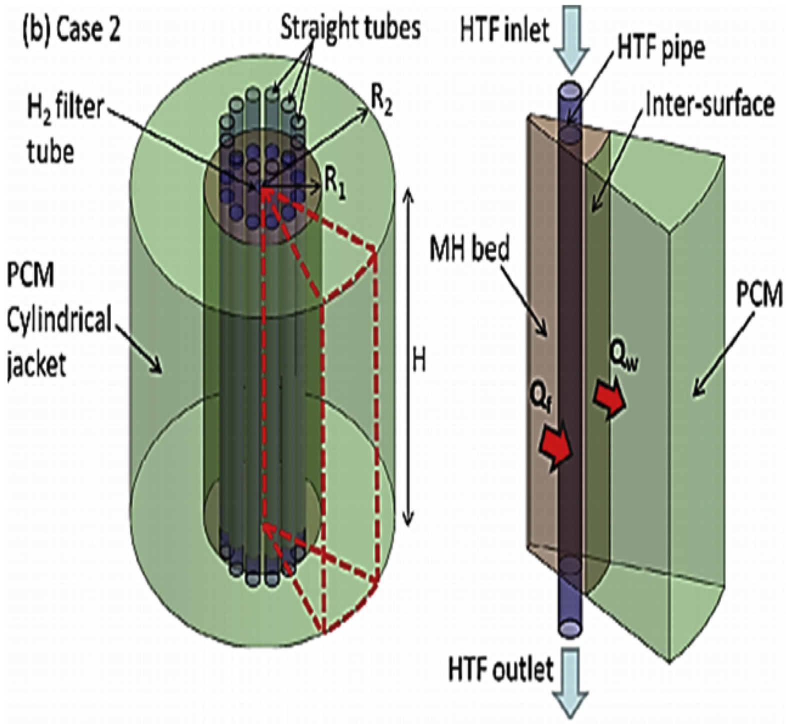
(a)



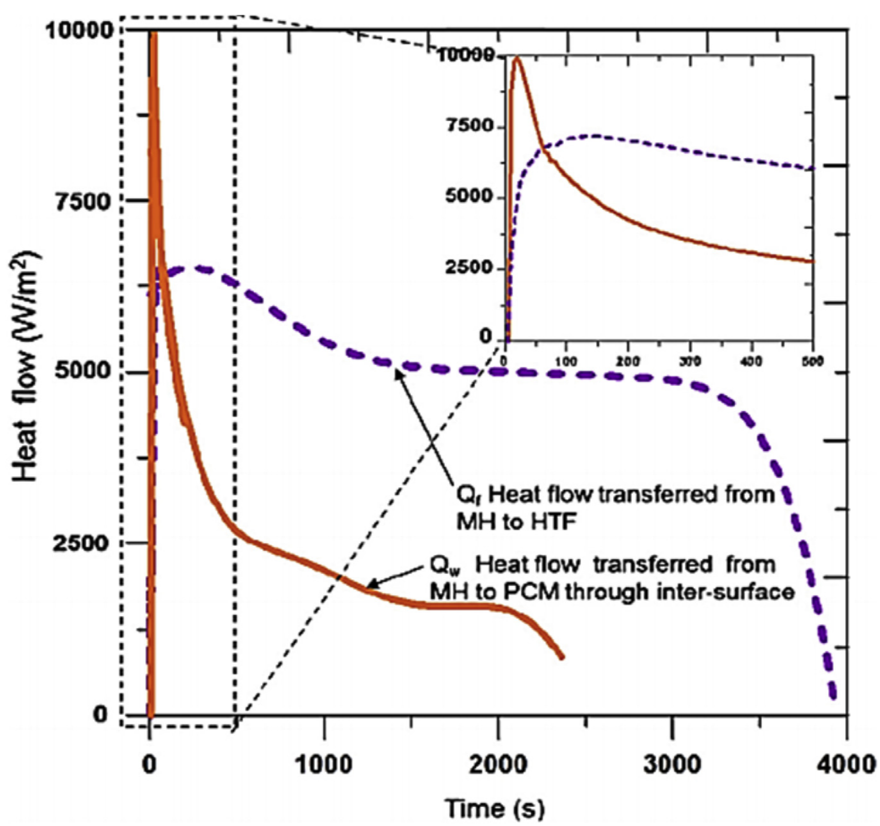
(b)

Fig. 9 – (a) Heat pipes integrated with PEMFC [102] (b) Metal Hydride tank with PCM [106].

use of PCM, it can be noted that addition of PCM increases the system volume and weight by nearly fifty per cent. Mellouli et al. numerically analyzed the performance of four different PCM based reactor designs [108]. The reactors comprise of spherical PCM shells, hexagonal tubes, cylindrical tubes and a single PCM tube placed at the center of the reactor. Cylindrical tubes displayed a better performance as compared to other configurations. They also integrated a HTF arrangement with a PCM jacket [109]. In the initial phase of the reaction, the PCM showed a high and rapid absorption of heat released much higher than the HTF; however, the peak lasted for less than 1% of the total time. For the most part HTF was the more active heat absorber, as can be seen from Fig. 10.



(a)



(b)

Fig. 10 — a) Metal hydride tank with PCM and HTF tubes b) Heat transfer to PCM and HTF [109].

Conclusions

While there are multiple ways of reaching an optimum design solution for metal hydride based hydrogen storage, the methodology is still evolving. A unified design philosophy that could at least define the most important parameters is lacking. For the most part of the process, heat transfer has been identified as the major limiting factor. The heat transfer methods generally used were fins, cooling tubes, metal foam, ENG and water jackets. Of these more than fifty per cent of the studies were on fins and cooling tubes. Thermal conductivity of the metal hydride was seen to be a major bottleneck for the performance of the storage tank. However, for reactors of a small diameter (less than 10 mm), an internal heat transfer enhancement may not be necessary. Metal foam was observed to be a more efficient heat transfer enhancement technique than fins, whereas ENG composites exhibited a better performance than metal foam and were also cost-effective. There are two aspects of heat withdrawal in a metal hydride reactor. First, the heat generated at the sites of reaction has to be dissipated through the system; this is where thermal conductivity of the metal hydride bed comes into play. Second, the heat dissipated through the metal hydride bed has to be removed via an external cooling arrangement such as water jacket, cooling tubes etc. It is the opinion of this review that any thermal management system that seeks to provide a solution to the heat transfer problem at hand should focus on improving both these factors simultaneously, as each is limited by the other. Therefore, an integration of multiple techniques is recommended such as a combination of mesh and cooling tubes or a combination of ENG and coiled tube etc. Improvement of only thermal conductivity or heat transfer coefficient or area may not be able to solve the heat transfer bottleneck.

REFERENCES

[1] Schlapbach L, Züttel A. Hydrogen-storage materials for mobile applications. Nature 2001;414:353–8. https://doi.org/10.1038/35104634.

[2] Huot J. Metal hydrides. In: Hirscher M, editor. Handb. hydrog. storage. Weinheim: Wiley-VCH; 2009. p. 81–116. https://doi.org/10.1002/9783527629800.

[3] Suda S, Kobayashi N, Morishita E, Takemoto N. Heat transmission analysis of metal hydride beds. J Less Common Met 1983;89:325–32. https://doi.org/10.1016/0022-5088(83)90341-7.

[4] Sun DW, Deng SJ. A theoretical model predicting the effective thermal conductivity in powdered metal hydride beds. Int J Hydrogen Energy 1990;15:331–6. https://doi.org/10.1016/0360-3199(90)90181-W.

[5] Sandrock G. Hydride storage. In: Vielstich W, Lamm A, Gasteiger HA, Yokokawa H, editors. Handb. Fuel Cells Fundam. Technol. Appl. New York: John Wiley and Sons; 2010. p. 1–12.

[6] Zhang J, Fisher TS, Ramachandran PV, Gore JP, Mudawar I. A review of heat transfer issues in hydrogen storage technologies. J Heat Transf 2005;127:1391–9. https://doi.org/10.1115/1.2098875.

[7] Yang FS, Wang GX, Zhang ZX, Meng XY, Rudolph V. Design of the metal hydride reactors – a review on the key technical issues. Int J Hydrogen Energy 2010;35:3832–40. https://doi.org/10.1016/j.ijhydene.2010.01.053.

[8] Mazzucco A, Dornheim M, Sloth M, Jensen TR, Jensen JO, Rokni M. Bed geometries, fueling strategies and optimization of heat exchanger designs in metal hydride storage systems for automotive applications: a review. Int J Hydrogen Energy 2014;39:17054–74. https://doi.org/10.1016/j.ijhydene.2014.08.047.

[9] Shafiee S, McCay MH. Different reactor and heat exchanger configurations for metal hydride hydrogen storage systems – a review. Int J Hydrogen Energy 2016;41:1–9. https://doi.org/10.1016/j.ijhydene.2016.03.133.

[10] Ben Nasrallah S, Jemni A. Heat and mass transfer models in metal-hydrogen reactor. Int J Hydrogen Energy 1997;22:67–76. https://doi.org/10.1016/S0360-3199(96)00039-0.

[11] Askri F, Jemni A, Ben Nasrallah S. Dynamic behavior of metal-hydrogen reactor during hydriding process. Int J Hydrogen Energy 2004;29:635–47. https://doi.org/10.1016/S0360-3199(03)00220-9.

[12] Askri F, Jemni A, Ben Nasrallah S. Prediction of transient heat and mass transfer in a closed metal hydrogen reactor. Int J Hydrogen Energy 2004;29:195–208. https://doi.org/10.1016/S0360-3199(03)00089-2.

[13] Chung CA, Ho CJ. Thermal-fluid behavior of the hydriding and dehydriding processes in a metal hydride hydrogen storage canister. Int J Hydrogen Energy 2009;34:4351–64. https://doi.org/10.1016/j.ijhydene.2009.03.028.

[14] Mellouli S, Askri F, Dhaou H, Jemni A, Ben Nasrallah S. Numerical study of heat exchanger effects on charge/discharge times of metal-hydrogen storage vessel. Int J Hydrogen Energy 2009;34:3005–17. https://doi.org/10.1016/j.ijhydene.2008.12.099.

[15] Nishizaki T, Miyamoto K, Yoshida K. Coefficients of performance of hydride heat pumps. J Less-Common Met 1983;89:559–66. https://doi.org/10.1016/0022-5088(83)90372-7.

[16] Choi H, Mills AF. Heat and mass transfer in metal hydride beds for heat pump applications. Int J Heat Mass Transf 1990;33:1281–8. https://doi.org/10.1016/0017-9310(90)90257-U.

[17] Gopal MR, Murthy SS. Prediction of heat and mass transfer in annular cylindrical metal hydride beds. Int J Hydrogen Energy 1992;17:795–805. https://doi.org/10.1016/0360-3199(92)90024-Q.

[18] Gopal MR, Murthy SS. Parametric studies on heat and mass transfer in metal hydride beds. Chem Eng Process Process Intensif 1993;32:217–23. https://doi.org/10.1016/0255-2701(93)80003-Y.

[19] Gopal MR, Murthy SS. Studies on heat and mass transfer in metal hydride beds. Int J Hydrogen Energy 1995;20:911–7. https://doi.org/10.1016/0360-3199(95)00026-A.

[20] Nagel M, Komazaki Y, Suda S. Effective thermal conductivity of a metal hydride bed augmented with a copper wire matrix. J Less Common Met 1986;120:35–43. https://doi.org/10.1016/0022-5088(86)90625-9.

[21] Chen Y, Sequeira CAC, Chen C, Wang X, Wang Q. Metal hydride beds and hydrogen supply tanks as minitype PEMFC hydrogen sources. Int J Hydrogen Energy 2003;28:329–33. https://doi.org/10.1016/S0360-3199(02)00064-2.

[22] Laurencelle F, Goyette J. Simulation of heat transfer in a metal hydride reactor with aluminium foam. Int J Hydrogen Energy 2007;32:2957–64. https://doi.org/10.1016/j.ijhydene.2006.12.007.

[23] Raju M, Kumar S. System simulation modeling and heat transfer in sodium alanate based hydrogen storage systems. Int J Hydrogen Energy 2011;36:1578–91. https://doi.org/10.1016/j.ijhydene.2010.10.100.

[24] Freni A, Cipitì F, Cacciola G. Finite element-based simulation of a metal hydride-based hydrogen storage tank. Int J Hydrogen Energy 2009;34:8574–82. https://doi.org/10.1016/j.ijhydene.2009.07.118.

[25] Wang H, Prasad AK, Advani SG. Hydrogen storage systems based on hydride materials with enhanced thermal conductivity. Int J Hydrogen Energy 2012;37:290–8. https://doi.org/10.1016/j.ijhydene.2011.04.096.

[26] Ferekh S, Gwak G, Kyoung S, Kang H, Chang M, Yun S, et al. Numerical comparison of heat-fin- and metal-foam-based hydrogen storage beds during hydrogen charging process. Int J Hydrogen Energy 2015;40:14540–50. https://doi.org/10.1016/j.ijhydene.2015.07.149.

[27] Madaria Y, Anil Kumar E, Srinivasa Murthy S. Effective thermal conductivity of reactive packed beds of hydriding materials. Appl Therm Eng 2016;98:976–90. https://doi.org/10.1016/j.applthermaleng.2016.01.013.

[28] Ron M, Gruen D, Mendelsohn M, Sheft I. Preparation and properties of porous metal hydride compacts. J Less Common Met 1980;74:445–8. https://doi.org/10.1016/0022-5088(80)90183-6.

[29] Tuscher E, Weinzierl P, Eder OJ. Porous metal hydride compacts: preparation, properties and use. Int J Hydrogen Energy 1983;8:199–203. https://doi.org/10.1016/0360-3199(83)90065-4.

[30] Ron M, Bershadsky E, Josephy Y. Thermal conductivity of PMH compacts, measurements and evaluation. Int J Hydrogen Energy 1992;17:623–30. https://doi.org/10.1016/0360-3199(92)90076-9.

[31] Lloyd G, Kim KJ, Razani A, Feldman TK. Thermal conductivity measurements of metal hydride compacts developed for high-power reactors. J Thermophys Heat Transf 1998;12:132–7. https://doi.org/10.2514/2.6332.

[32] Sánchez AR, Klein H-P, Groll M. Expanded graphite as heat transfer matrix in metal hydride beds. Int J Hydrogen Energy 2003;28:515–27. https://doi.org/10.1016/S0360-3199(02)00057-5.

[33] Pohlmann C, Röntzsch L, Kalinichenka S, Hutsch T, Kieback B. Magnesium alloy-graphite composites with tailored heat conduction properties for hydrogen storage applications. Int J Hydrogen Energy 2010;35:12829–36. https://doi.org/10.1016/j.ijhydene.2010.08.104.

[34] Pohlmann C, Röntzsch L, Weißgärber T, Kieback B. Heat and gas transport properties in pelletized hydride-graphite-composites for hydrogen storage applications. Int J Hydrogen Energy 2013;38:1685–91. https://doi.org/10.1016/j.ijhydene.2012.09.159.

[35] Dieterich M, Pohlmann C, Bürger I, Linder M, Röntzsch L. Long-term cycle stability of metal hydride-graphite composites. Int J Hydrogen Energy 2015;40:16375–82. https://doi.org/10.1016/j.ijhydene.2015.09.013.

[36] von Colbe JMB, Lozano G, Metz O, Bücherl T, Bormann R, Klassen T, et al. Design, sorption behaviour and energy management in a sodium alanate-based lightweight hydrogen storage tank. Int J Hydrogen Energy 2015;40:2984–8. https://doi.org/10.1016/j.ijhydene.2015.01.013.

[37] Nishizaki T, Miyamoto M, Miyamoto K, Yoshida K, Yamaji K, Nakata Y. Metal hydride reactor. U.S. Patent 4,457,136, 1984.

[38] Guo Z. Enhancement of heat and mass transfer in metal hydride beds with the addition of Al plates. Heat Mass Transf 1999;34:517–23. https://doi.org/10.1007/s002310050290.

[39] Oi T, Maki K, Sakaki Y. Heat transfer characteristics of the metal hydride vessel based on the plate-fin type heat exchanger. J Power Sources 2004;125:52–61. https://doi.org/10.1016/S0378-7753(03)00822-X.

[40] Botzung M, Chaudourne S, Gillia O, Perret C, Latroche M, Percheron-Guegan A, et al. Simulation and experimental validation of a hydrogen storage tank with metal hydrides. Int J Hydrogen Energy 2008;33:98–104. https://doi.org/10.1016/j.ijhydene.2007.08.030.

[41] MacDonald BD, Rowe AM. Impacts of external heat transfer enhancements on metal hydride storage tanks. Int J Hydrogen Energy 2006;31:1721–31. https://doi.org/10.1016/j.ijhydene.2006.01.007.

[42] Veerraju C, Gopal MR. Heat and mass transfer studies on elliptical metal hydride tubes and tube banks. Int J Hydrogen Energy 2009;34:4340–50. https://doi.org/10.1016/j.ijhydene.2009.03.022.

[43] Veerraju C, Ram Gopal M. Heat and mass transfer studies on plate fin-and-elliptical tube type metal hydride reactors. Appl Therm Eng 2010;30:673–82. https://doi.org/10.1016/j.applthermaleng.2009.11.015.

[44] Visaria M, Mudawar I, Pourpoint T. Enhanced heat exchanger design for hydrogen storage using high-pressure metal hydride: part 1. Design methodology and computational results. Int J Heat Mass Transf 2011;54:413–23. https://doi.org/10.1016/j.ijheatmasstransfer.2010.09.029.

[45] Andreasen G, Melnichuk M, Ramos S, Corso HL, Visintin A, Triaca WE, et al. Hydrogen desorption from a hydride container under different heat exchange conditions. Int J Hydrogen Energy 2013;38:13352–9. https://doi.org/10.1016/j.ijhydene.2013.07.115.

[46] Satya Sekhar B, Lototskyy M, Kolesnikov A, Moropeng ML, Tarasov BP, Pollet BG. Performance analysis of cylindrical metal hydride beds with various heat exchange options. J Alloys Compd 2015;645:S89–95. https://doi.org/10.1016/j.jallcom.2014.12.272.

[47] Askri F, Ben Salah M, Jemni a, Ben Nasrallah S. Optimization of hydrogen storage in metal-hydride tanks. Int J Hydrogen Energy 2009;34:897–905. https://doi.org/10.1016/j.ijhydene.2008.11.021.

[48] Kim MC, Lee SY, Gu JH. Numerical prediction of hydrogen storaging performance of finned metal hybride beds. Trans Korean Soc Mech Eng B 1998;22:520–9. https://doi.org/10.22634/KSME-B.1998.22.4.520.

[49] Melnichuk M, Silin N, Peretti HA. Optimized heat transfer fin design for a metal-hydride hydrogen storage container. Int J Hydrogen Energy 2009;34:3417–24. https://doi.org/10.1016/j.ijhydene.2009.02.040.

[50] Ma J, Wang Y, Shi S, Yang F, Bao Z, Zhang Z. Optimization of heat transfer device and analysis of heat & mass transfer on the finned multi-tubular metal hydride tank. Int J Hydrogen Energy 2014;39:13583–95. https://doi.org/10.1016/j.ijhydene.2014.03.016.

[51] Singh A, Maiya MP, Murthy SS. Effects of heat exchanger design on the performance of a solid state hydrogen storage device. Int J Hydrogen Energy 2015;40:9733–46. https://doi.org/10.1016/j.ijhydene.2015.06.015.

[52] Garrier S, Chaise A, De Rango P, Marty P, Delhomme B, Fruchart D, et al. MgH_2 intermediate scale tank tests under various experimental conditions. Int J Hydrogen Energy 2011;36:9719–26. https://doi.org/10.1016/j.ijhydene.2011.05.017.

[53] Delhomme B, De Rango P, Marty P, Bacia M, Zawilski B, Raufast C, et al. Large scale magnesium hydride tank coupled with an external heat source. Int J Hydrogen Energy 2012;37:9103–11. https://doi.org/10.1016/j.ijhydene.2012.03.018.

[54] Nyamsi SN, Yang F, Zhang Z. An optimization study on the finned tube heat exchanger used in hydride hydrogen storage system - analytical method and numerical simulation. Int J Hydrogen Energy 2012;37:16078–92. https://doi.org/10.1016/j.ijhydene.2012.08.074.
[55] Kukkapalli VK, Kim SW. Metal hydride reactor design optimization for hydrogen energy storage. Key Eng Mater 2016;708:85–93. https://doi.org/10.4028/www.scientific.net/KEM.708.85.
[56] Sun DW. Designs of metal hydride reactors. Int J Hydrogen Energy 1992;17:945–9. https://doi.org/10.1016/0360-3199(92)90055-2.
[57] Kikkinides ES, Georgiadis MC, Stubos AK. Dynamic modelling and optimization of hydrogen storage in metal hydride beds. Energy 2006;31:2092–110. https://doi.org/10.1016/j.energy.2005.10.036.
[58] Souahlia A, Dhaou H, Askri F, Mellouli S, Jemni A, Ben Nasrallah S. Experimental study and characterization of metal hydride containers. Int J Hydrogen Energy 2011;36:4952–7. https://doi.org/10.1016/j.ijhydene.2011.01.074.
[59] Souahlia A, Dhaou H, Askri F, Sofiene M, Jemni A, Ben Nasrallah S. Experimental and comparative study of metal hydride hydrogen tanks. Int J Hydrogen Energy 2011;36:12918–22. https://doi.org/10.1016/j.ijhydene.2011.07.022.
[60] Raju M, Kumar S. Optimization of heat exchanger designs in metal hydride based hydrogen storage systems. Int J Hydrogen Energy 2012;37:2767–78. https://doi.org/10.1016/j.ijhydene.2011.06.120.
[61] Bao Z. Performance investigation and optimization of metal hydride reactors for high temperature thermochemical heat storage. Int J Hydrogen Energy 2015;40:5664–76. https://doi.org/10.1016/j.ijhydene.2015.02.123.
[62] Mazzucco A, Voskuilen TG, Waters EL, Pourpoint TL, Rokni M. Heat exchanger selection and design analyses for metal hydride heat pump systems. Int J Hydrogen Energy 2016;41:4198–213. https://doi.org/10.1016/j.ijhydene.2016.01.016.
[63] Kaplan Y. Effect of design parameters on enhancement of hydrogen charging in metal hydride reactors. Int J Hydrogen Energy 2009;34:2288–94. https://doi.org/10.1016/j.ijhydene.2008.12.096.
[64] Na Ranong C, Höhne M, Franzen J, Hapke J, Fieg G, Dornheim M, et al. Concept, design and manufacture of a prototype hydrogen storage tank based on sodium alanate. Chem Eng Technol 2009;32:1154–63. https://doi.org/10.1002/ceat.200900095.
[65] Lozano GA, Ranong CN, Bellosta Von Colbe JM, Bormann R, Hapke J, Fieg G, et al. Optimization of hydrogen storage tubular tanks based on light weight hydrides. Int J Hydrogen Energy 2012;37:2825–34. https://doi.org/10.1016/j.ijhydene.2011.03.043.
[66] Mohan G, Prakash Maiya M, Srinivasa Murthy S. Performance simulation of metal hydride hydrogen storage device with embedded filters and heat exchanger tubes. Int J Hydrogen Energy 2007;32:4978–87. https://doi.org/10.1016/j.ijhydene.2007.08.007.
[67] Boukhari A, Bessaih R. Numerical heat and mass transfer investigation of hydrogen absorption in an annulus-disc reactor. Int J Hydrogen Energy 2015;40:13708–17. https://doi.org/10.1016/j.ijhydene.2015.05.123.
[68] Bao Z, Yang F, Wu Z, Nyamsi SN, Zhang Z. Optimal design of metal hydride reactors based on CFD-Taguchi combined method. Energy Convers Manag 2013;65:322–30. https://doi.org/10.1016/j.enconman.2012.07.027.
[69] Muthukumar P, Singhal A, Bansal GK. Thermal modeling and performance analysis of industrial-scale metal hydride based hydrogen storage container. Int J Hydrogen Energy 2012;37:14351–64. https://doi.org/10.1016/j.ijhydene.2012.07.010.
[70] Garrison SL, Hardy BJ, Gorbounov MB, Tamburello DA, Corgnale C, Vanhassel BA, et al. Optimization of internal heat exchangers for hydrogen storage tanks utilizing metal hydrides. Int J Hydrogen Energy 2012;37:2850–61. https://doi.org/10.1016/j.ijhydene.2011.07.044.
[71] Gkanas EI, Grant DM, Khzouz M, Stuart AD, Manickam K, Walker GS. Efficient hydrogen storage in up-scale metal hydride tanks as possible metal hydride compression agents equipped with aluminium extended surfaces. Int J Hydrogen Energy 2016;41:10795–810. https://doi.org/10.1016/j.ijhydene.2016.04.035.
[72] Dhaou H, Souahlia A, Mellouli S, Askri F, Jemni A, Ben Nasrallah S. Experimental study of a metal hydride vessel based on a finned spiral heat exchanger. Int J Hydrogen Energy 2010;35:1674–80. https://doi.org/10.1016/j.ijhydene.2009.11.094.
[73] Mellouli S, Askri F, Dhaou H, Jemni A, Ben Nasrallah S. Numerical simulation of heat and mass transfer in metal hydride hydrogen storage tanks for fuel cell vehicles. Int J Hydrogen Energy 2010;35:1693–705. https://doi.org/10.1016/j.ijhydene.2009.12.052.
[74] Flueckiger S, Voskuilen T, Pourpoint T, Fisher TS, Zheng Y. In situ characterization of metal hydride thermal transport properties. Int J Hydrogen Energy 2010;35:614–21. https://doi.org/10.1016/j.ijhydene.2009.11.001.
[75] Visaria M, Mudawar I, Pourpoint T, Kumar S. Study of heat transfer and kinetics parameters influencing the design of heat exchangers for hydrogen storage in high-pressure metal hydrides. Int J Heat Mass Transf 2010;53:2229–39. https://doi.org/10.1016/j.ijheatmasstransfer.2009.12.010.
[76] Visaria M, Mudawar I, Pourpoint T. Enhanced heat exchanger design for hydrogen storage using high-pressure metal hydride - part 2. Experimental results. Int J Heat Mass Transf 2011;54:424–32. https://doi.org/10.1016/j.ijheatmasstransfer.2010.09.028.
[77] Visaria M, Mudawar I. Coiled-tube heat exchanger for high-pressure metal hydride hydrogen storage systems - part 2. Computational model. Int J Heat Mass Transf 2012;55:1796–806. https://doi.org/10.1016/j.ijheatmasstransfer.2011.11.036.
[78] Visaria M, Mudawar I. Coiled-tube heat exchanger for high-pressure metal hydride hydrogen storage systems - part 1. Experimental study. Int J Heat Mass Transf 2012;55:1782–95. https://doi.org/10.1016/j.ijheatmasstransfer.2011.11.035.
[79] Wu Z, Yang F, Zhang Z, Bao Z. Magnesium based metal hydride reactor incorporating helical coil heat exchanger: simulation study and optimal design. Appl Energy 2014;130:712–22. https://doi.org/10.1016/j.apenergy.2013.12.071.
[80] Nakano A, Maeda T, Ito H, Motyka T, Perez-Berrios JM, Greenway S. Experimental study on a metal hydride tank for the totalized hydrogen energy utilization system. Energy Procedia 2012;29:463–8. https://doi.org/10.1016/j.egypro.2012.09.054.
[81] Nakano A, Ito H, Maeda T, Munakata T, Motyka T, Corgnale C, et al. Study on a metal hydride tank to support energy storage for renewable energy. J Alloys Compd 2013;580:S418–22. https://doi.org/10.1016/j.jallcom.2013.03.152.
[82] Muthukumar P, Maiya MP, Murthy SS. Experiments on a metal hydride-based hydrogen storage device. Int J Hydrogen Energy 2005;30:1569–81. https://doi.org/10.1016/j.ijhydene.2004.12.007.
[83] Muthukumar P, Madhavakrishna U, Dewan A. Parametric studies on a metal hydride based hydrogen storage device. Int J Hydrogen Energy 2007;32:4988–97. https://doi.org/10.1016/j.ijhydene.2007.08.010.

[84] Jiao K, Li X, Yin Y, Zhou Y, Yu S, Du Q. Effects of various operating conditions on the hydrogen absorption processes in a metal hydride tank. Appl Energy 2012;94:257–69. https://doi.org/10.1016/j.apenergy.2012.01.033.
[85] Ye J, Jiang L, Li Z, Liu X, Wang S, Li X. Numerical analysis of heat and mass transfer during absorption of hydrogen in metal hydride based hydrogen storage tanks. Int J Hydrogen Energy 2010;35:8216–24. https://doi.org/10.1016/j.ijhydene.2009.12.086.
[86] Krokos CA, Nikolic D, Kikkinides ES, Georgiadis MC, Stubos AK. Modeling and optimization of multi-tubular metal hydride beds for efficient hydrogen storage. Int J Hydrogen Energy 2009;34:9128–40. https://doi.org/10.1016/j.ijhydene.2009.09.021.
[87] Gambini M, Manno M, Vellini M. Numerical analysis and performance assessment of metal hydride-based hydrogen storage systems. Int J Hydrogen Energy 2008;33:6178–87. https://doi.org/10.1016/j.ijhydene.2008.08.006.
[88] Bhouri M, Goyette J, Hardy BJ, Anton DL. Numerical modeling and performance evaluation of multi-tubular sodium alanate hydride finned reactor. Int J Hydrogen Energy 2012;37:1551–67. https://doi.org/10.1016/j.ijhydene.2011.10.044.
[89] Bao Z, Yang F, Wu Z, Cao X, Zhang Z. Simulation studies on heat and mass transfer in high-temperature magnesium hydride reactors. Appl Energy 2013;112:1181–9. https://doi.org/10.1016/j.apenergy.2013.04.053.
[90] Anbarasu S, Muthukumar P, Mishra SC. Tests on $LmNi_{4.91}Sn_{0.15}$ based solid state hydrogen storage device with embedded cooling tubes - part B: desorption process. Int J Hydrogen Energy 2014;39:4966–72. https://doi.org/10.1016/j.ijhydene.2014.01.039.
[91] Anbarasu S, Muthukumar P, Mishra SC. Tests on $LmNi_{4.91}Sn_{0.15}$ based solid state hydrogen storage device with embedded cooling tubes - part A: absorption process. Int J Hydrogen Energy 2014;39:3342–51. https://doi.org/10.1016/j.ijhydene.2013.12.090.
[92] Valizadeh M, Delavar MA, Farhadi M. Numerical simulation of heat and mass transfer during hydrogen desorption in metal hydride storage tank by Lattice Boltzmann method. Int J Hydrogen Energy 2016;41:413–24. https://doi.org/10.1016/j.ijhydene.2015.11.075.
[93] Linder M, Mertz R, Laurien E. Experimental analysis of fast metal hydride reaction bed dynamics. Int J Hydrogen Energy 2010;35:8755–61. https://doi.org/10.1016/j.ijhydene.2010.05.023.
[94] Yang F, Meng X, Deng J, Wang Y, Zhang Z. Identifying heat and mass transfer characteristics of metal hydride reactor during adsorption-parameter analysis and numerical study. Int J Hydrogen Energy 2008;33:1014–22. https://doi.org/10.1016/j.ijhydene.2007.11.007.
[95] Yang F, Meng X, Deng J, Wang Y, Zhang Z. Identifying heat and mass transfer characteristics of metal hydride reactor during adsorption: improved formulation about parameter analysis. Int J Hydrogen Energy 2009;34:1852–61. https://doi.org/10.1016/j.ijhydene.2008.11.119.
[96] Dunikov DO, Borzenko VI, Malyshenko SP. Optimization of heat transfer in metal hydride reactor. In: Proc WHEC, Energy Environ, vol. 78; 2010. p. 88–94. Essen.
[97] Borzenko VI, Dunikov DO, Malyshenko SP. Crisis phenomena in metal hydride hydrogen storage facilities. High Temp 2011;49:249–56. https://doi.org/10.1134/S0018151X11010019.
[98] Herbrig K, Röntzsch L, Pohlmann C, Weißgärber T, Kieback B. Hydrogen storage systems based on hydride-graphite composites: computer simulation and experimental validation. Int J Hydrogen Energy 2013;38:7026–36. https://doi.org/10.1016/j.ijhydene.2013.03.104.
[99] Førde T, Eriksen J, Pettersen AG, Vie PJS, Ulleberg. Thermal integration of a metal hydride storage unit and a PEM fuel cell stack. Int J Hydrogen Energy 2009;34:6730–9. https://doi.org/10.1016/j.ijhydene.2009.05.146.
[100] Chung CA, Yang SW, Yang CY, Hsu CW, Chiu PY. Experimental study on the hydrogen charge and discharge rates of metal hydride tanks using heat pipes to enhance heat transfer. Appl Energy 2013;103:581–7. https://doi.org/10.1016/j.apenergy.2012.10.024.
[101] Chung CA, Chen YZ, Chen YP, Chang MS. CFD investigation on performance enhancement of metal hydride hydrogen storage vessels using heat pipes. Appl Therm Eng 2015;91:434–46. https://doi.org/10.1016/j.applthermaleng.2015.08.039.
[102] Tetuko AP, Shabani B, Andrews J. Thermal coupling of PEM fuel cell and metal hydride hydrogen storage using heat pipes. Int J Hydrogen Energy 2016;41:4264–77. https://doi.org/10.1016/j.ijhydene.2015.12.194.
[103] Meng X, Wu Z, Bao Z, Yang F, Zhang Z. Performance simulation and experimental confirmation of a mini-channel metal hydrides reactor. Int J Hydrogen Energy 2013;38:15242–53. https://doi.org/10.1016/j.ijhydene.2013.09.056.
[104] Mudawwar I, Visaria M. Coiled and microchannel heat exchangers for metal hydride storage systems. US 8,778,063 B2. 2014. https://doi.org/10.1016/j. (73).
[105] Garrier S, Delhomme B, De Rango P, Marty P, Fruchart D, Miraglia S. A new MgH_2 tank concept using a phase-change material to store the heat of reaction. Int J Hydrogen Energy 2013;38:9766–71. https://doi.org/10.1016/j.ijhydene.2013.05.026.
[106] Darzi AAR, Afrouzi HH, Moshfegh A, Farhadi M. Absorption and desorption of hydrogen in long metal hydride tank equipped with phase change material jacket. Int J Hydrogen Energy 2016;41:9595–610. https://doi.org/10.1016/j.ijhydene.2016.04.051.
[107] Ben Mâad H, Askri F, Ben Nasrallah S. Heat and mass transfer in a metal hydrogen reactor equipped with a phase-change heat-exchanger. Int J Therm Sci 2016;99:271–8. https://doi.org/10.1016/j.ijthermalsci.2015.09.003.
[108] Mellouli S, Abhilash E, Askri F, Ben Nasrallah S. Integration of thermal energy storage unit in a metal hydride hydrogen storage tank. Appl Therm Eng 2016;102:1185–96. https://doi.org/10.1016/j.applthermaleng.2016.03.116.
[109] Mellouli S, Askri F, Abhilash E, Ben Nasrallah S. Impact of using a heat transfer fluid pipe in a metal hydride-phase change material tank. Appl Therm Eng 2017;113:554–65. https://doi.org/10.1016/j.applthermaleng.2016.11.065.